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STRATEGIC DIRECTIONS FOR AI: THE ROLE OF CIOS AND BOARDS OF DIRECTORS[1]

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This paper applies upper echelons theory to investigate whether chief information officers (CIOs) and boards of directors affect the development of AI orientation, which represents firms' overall strategic direction and goals regarding the introduction and application of artificial intelligence (AI) technology. We tested our model using a dataset drawn from 1,454 publicly listed firms in China. Our findings show that the presence of a CIO positively influences AI orientation and that board educational diversity, R&D experience, and AI experience positively moderate the CIO's effect on AI orientation. Our post hoc analysis further demonstrates that these board characteristics represent contingencies that impact AI orientation but not conventional IT orientation. This paper contributes to the upper echelons literature and IT management research by offering contextualized arguments that explain new business and IT strategies such as AI orientation. Further, our findings suggest important implications about how to build top management teams and boards capable of effectively developing AI orientations.

Keywords: AI orientation, chief information officer (CIO), board of directors, board diversity, upper echelons theory, public firms

Introduction

As a disruptive innovation, artificial intelligence (AI) has the potential to force firms to restructure or redefine their business models in order to survive, sustain operations, and gain competitive advantages (Davenport et al. 2020). Defined as technologies that leverage machine-based intelligence and advanced computing capacity to mimic human "cognitive" functions, AI goes beyond conventional technologies by approximating human cognitive functions to search, analyze, and make decisions based on large-scale data (Davenport et al. 2020; Simon 1995). AI has wide-reaching implications for

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firm production and operations because it can change how firms make decisions and offers opportunities for mass customization (Davenport and Ronanki 2018). To realize such opportunities, AI requires firms to develop a distinct technological environment that requires substantial investments and corresponding changes in personnel and business policies.

While AI offers tremendous opportunities for firms, it also presents risks. For example, Under Armour, a sports and clothing company, used an AI-based cognitive computing platform to transform health and fitness insights, realizing a 51% revenue increase (about $80 million USD) (Ratnayake 2020). However, unsuccessful cases of AI deployment have resulted in substantial losses. For instance, after 4 years and a 62 million USD investment, the Anderson Cancer Center failed to develop an effective AI-based cancer diagnosis system (Jaklevic 2017). Because of its significant potential upsides and downsides, developing a strategic position to effectively manage AI is an important challenge for top managers (Diorio 2020).

Navigating the opportunities and risks associated with AI implementation requires firms to develop a strategic AI orientation. Defined as *a firm's overall strategic direction and goals associated with introducing and applying AI technology*, AI orientation provides managers with a framework to guide how they invest, manage, and apply it within their firms (Miles and Arnold 1991; Peterson 1989). By developing an AI orientation, firms can maximize the benefits of AI while minimizing the associated costs and risks (McKinsey & Company 2019). Upper echelons theory suggests that the presence of certain top management roles can significantly affect firm strategies and outcomes. For example, the presence of chief sustainability officers (CSOs) reduces socially irresponsible behaviors (Fu et al. 2020). Therefore, in order to understand how firms develop AI orientations, we examine whether firms have a chief information officer (CIO), as this role has the capacity to shape top management's understanding of AI's potential as an innovative technology, as well as its implications for overall strategy (Chatterjee et al. 2001).

Because CIOs are likely to understand the strategic value of new technologies and possess experience aligning IT-related decisions with business goals and orientations, they hold a key role in the development of a firm's AI orientation (Gerow et al. 2014; Karahanna and Preston 2013). Much work on CIOs and strategy has directed attention to conventional IT contexts, evaluating overall IT spending, especially in markets where CIOs' presence is pervasive (e.g., the United States). However, the specific role that CIOs play in shaping new IT strategies remains poorly understood—particularly in emerging markets where IT leaders are not well-represented in top management teams. For example, there are distinct differences in terms of CIO presence in the Chinese and U.S. markets: whereas 4.6% of S&P 1500 firms listed on the ExecuComp database had a formal CIO or chief technology officer (CTO), only 1.8% of the 3,600 public firms listed on the China Stock Market & Accounting Research Database (CSMAR) in 2018 had a formal CIO role. Therefore, particularly in emerging markets, it is necessary to explore how CIO presence impacts AI orientation. We thus ask our first research question: *Can the presence of a CIO facilitate AI orientation in firms?*

To develop an effective AI orientation, CIOs need support from their boards of directors. Recent updates to upper echelons theory emphasize that boards can constrain the impact of top managers (Hambrick 2007; Hambrick et al. 2005; Hambrick and Mason 1984). Although prior IS research has shown that boards do not pay attention to IT (e.g., Huff et al. 2006), we argue that the opposite is true in the context of AI orientation. Indeed, the novel characteristics of AI and their potential to drive negative firm performance should evoke great interest among board members (Johnson et al. 1993; Tuggle et al. 2010). Further, deep applications of AI may change boards' work environments, job requirements, and power dynamics. Consequently, boards should be motivated to participate in crafting an AI orientation (Zhu and Yoshikawa 2016). Therefore, our second research question is: *How do boards affect the relationship between the CIO and AI orientation?*

To address these research questions, we build on upper echelons theory and propose that firms with CIOs are more likely to develop an AI orientation because the CIO provides the technological knowledge and understanding necessary to align AI with strategic business objectives. Further, we propose that boards with more educational diversity and R&D experience are better able to work with CIOs to increase a firm's likelihood of developing an AI orientation. Also, we argue that boards with members who have experience with AI orientations through interlock companies possess the knowledge necessary to help CIOs develop an AI orientation. We tested our hypotheses using a unique dataset drawn from the annual reports of 1,454 publicly listed firms. The results support our hypotheses.

This study makes several contributions. First, we contribute to the IT management literature by offering a strategic perspective on how upper echelons influence the development of an AI orientation, and demonstrate that CIOs interact with boards to shape AI orientation. We also confirm findings in previous literature that boards are unlikely to engage with conventional IT orientation. Second, we show that board educational diversity, R&D experience, and AI orientation experience moderate the relationship between the CIO and AI orientation. We thus offer a nuanced understanding of factors that shape boards' engagement with AI and empirically demonstrate that these factors do not affect conventional IT orientation. Third, we enrich the upper echelons literature by exploring whether and how the presence of a CIO affects AI orientation. By providing evidence supporting the strategic importance of CIO presence, our study underscores the practical importance of a firm appointing a CIO in emerging markets. Fourth, as one of the very few empirical studies investigating the interaction between managers and boards, this study provides a timely response to the call for research examining the interaction between CIOs and boards for enhancing firms' IT orientations in general, and AI orientations in particular (Cheng et al. 2021).

Theoretical Foundations and Hypotheses

AI and AI Orientation

AI leverages machine-based intelligence and advanced computing capacity to mimic human "cognitive" functions (Simon 1995). Typically, AI integrates technologies such as machine learning, deep learning, natural language processing, and computer vision to create high-intelligence algorithms that approximate human cognitive functions to search, analyze, and make decisions based on large-scale data (Davenport et al. 2020; Simon 1995). For example, Alibaba's AI algorithms not only process huge amounts of data but also make decisions and engage in direct action, continuously generating new decisions and recommendations by learning from customer reactions to previous decisions without human intervention. AI applications can help firms go beyond the constraints imposed by human cognition by creating novel insight into the structures, products, and processes necessary to sustain competitiveness (Ransbotham et al. 2018). For instance, Alibaba leverages AI technologies to predict what customers might want to buy and to automatically generate product descriptions for the e-commerce platform (Marr 2020). Thus, firms such as Alibaba can use AI to go beyond the limits of conventional IT and deliver products and services much more efficiently.

AI is distinct from conventional IT that enables acquiring and processing data, because AI applications can leverage data to perform automatic and enhanced self-learning actions and can make decisions on behalf of people and/or organizations (Faraj et al. 2018). Realizing AI's capacity to learn from data requires the transformation of firm technological and social structures. For example, training and sustaining self-learning algorithms requires continuous extraction of large volumes of high-quality data. Therefore, to support AI, firms need to establish a scalable processing system capable of generating and curating data from internal and external sources (Nemati et al. 2002).

Constructing an overall data environment demands that firms establish AI evaluation and development standards. Given that AI is a new technology, it is subject to few well-established quality evaluation standards (Rudin 2019). As a result, when introducing AI, it is essential that managers establish standards to evaluate the quality of decisions suggested by AI algorithms and assess the ability of AI to manage firm processes. Absent such standards, firms will face ambiguity about whether and when to rely on decisions recommended by AI (Makridakis 2017).

Establishing a scalable data environment to support AI requires firms to build and support human capital. Firms will need policies to clarify when supporting AI demands recruiting specific skill sets, when to replace current employees with those who have AI-specific IT skills, and when to retrain current employees to obtain or enhance AI-specific IT skills (Rana and Sharma 2019). Substantial risks associated with the structural changes necessary to develop and sustain such human capital demand the attention of top managers (Faraj et al. 2018).

Implementing AI makes developing standard operating procedures for identifying and mitigating failures necessary. Currently, most AI-based processes, such as machine learning and deep learning remain a "black box." While AI may outperform human decision makers in terms of speed or accuracy in analysis, it may also make suboptimal decisions caused by a lack of awareness of sociotechnical context; for example, when an AI chooses to source lower-priced goods from a problematic location. Therefore, standard operating procedures are needed to guide human decision makers in accepting or rejecting AI recommendations. Further, if AI recommendations lead to negative outcomes, standard operating procedures can help firms respond to failure and minimize losses. Given that there is scant knowledge of how to assess AI performance, such standard operating procedures can help identify uncertainties and reduce the potential costs of failure.

Deploying AI also requires top managers to define shared objectives and attitudes that align employee expectations with the planned impact of AI on business processes and structures. Especially given the frequent media reports that AI will eventually replace human workers in jobs ranging from the stock market floor to the factory floor (Kelly 2020), employees may experience greater uncertainty and perceive identity threats, resulting in resistance to the introduction of AI (Craig et al. 2019) unless they are presented with a clear understanding of the planned, positive impact of AI.

In summary, to manage the complexities of its deployment, we argue that it is essential for firms to develop a strategic orientation on AI. AI orientation, or *a firm's overall strategic direction and goals associated with introducing and applying AI technology*, guides AI-related strategic decisions, including AI investment, AI adoption, and AI-related management practices (Ding et al. 2014; Li, Wei, and Liu 2010). By articulating shared objectives, AI orientation facilitates managers' decisions regarding necessary investments, inspires managers to apply AI to solve business problems, and communicates the value and logic guiding the strategic use of AI to stakeholders. Thus, AI orientation can maximize the opportunities presented by introducing AI to firm processes while minimizing any associated costs and risks (McKinsey & Company 2019).

The Upper Echelons Theory and the CIO

Defined as top managers who are responsible for leading firm operations, upper echelons are responsible for developing a firm's AI orientation. As top decision makers, upper echelons are best positioned to understand the market position, resources, and capabilities that are most relevant to AI for the organization. Upper echelons theory suggests that organizational outcomes and strategic selections can be viewed as a function of upper echelon characteristics (Carpenter et al. 2004; Hambrick and Mason 1984). Specifically, strategic outcomes result from choices made by upper echelons that reflect their values, cognitions, and perceptions (Klein and Harrison 2007). Cognition can be measured through observable individual and group characteristics, functional tracks, career experiences, education, socioeconomic backgrounds, financial position, and group characteristics (Carpenter et al. 2004; Hambrick and Mason 1984; Klein and Harrison 2007).

Upper echelons have a strong motivation to participate in developing AI orientations because AI has the capacity to change the power dynamics and culture within a firm and can alter the dynamics of the top management team itself by changing the firm's resource allocation, procedures, and objectives. Top managers with less technological expertise may find it is difficult to understand the logic and processes that inform decisions made by AI, leading to feelings of confusion, uncertainty, and concern toward AI and motivating them to turn to upper echelon members with stronger technological expertise (Brock and von Wangenheim 2019). As a result, top managers with more technological expertise and experience, such as CIOs, CTOs, or CDOs (Chief Data Officers), may have more power to shape AI orientation and the associated standard operating procedures for compliance with AI recommendations (Singh and Hess 2017).

Since CIOs possess knowledge that is different from other upper echelon members and have distinct responsibilities in aligning IT with firm strategy, IS scholars have studied how CIOs affect firm-level outputs (see Appendix A). Among upper echelons, CIOs are most often responsible for identifying, purchasing, and implementing cutting-edge technologies (Weill and Woerner 2013). Beyond supporting overall IT operations, the CIO is the upper echelon member responsible for ensuring that an IT orientation aligns with a firm's general strategy (Grover et al. 2007). Specifically, the CIO is responsible for formulating new rules, promoting a technologically oriented culture, and developing norms that increase employees' recognition of and appreciation for advanced technologies (Liang et al. 2007). CIOs can also help improve technological capabilities and promote digital organizational transformation (Chun and Mooney 2009; Peppard et al. 2011; Weill and Woerner 2013) (see Appendix B).

CIOs are in a position to help other top managers understand the firm's technological capabilities and resources. CIOs can leverage their greater technological skills to increase technological investment and organizational performance (Carter et al. 2011; Sobol and Klein 2009) by promoting strategic alignment and firm performance (Enns et al. 2007; Karahanna and Preston 2013). CIOs commonly promote IS strategic alignment (Gerow et al. 2017; Karahanna and Preston 2013; Valorinta 2011; Wu et al. 2015) by shaping mutual understanding (Liang et al. 2017; Preston and Karahanna 2009), shared language and terminology, shared cognition, and trust among members of the upper echelon.

While studies have shed light on tactics employed by effective CIOs, further research is needed that directly examines the strategic implications of CIO presence on top management teams. Current upper echelons research assumes that CIOs are present, and typically focuses on CIO characteristics such as personal attributes, job contents, and reporting structures (Banker et al. 2011). However, evidence suggests that CIOs are not always present in top management teams. According to the ExecuComp database, as of 2018, only 4.6% of the 1,763 firms listed on the S&P 1500 Index in 2018 had formal CIO or CTO positions. Furthermore, CIO presence may be even rarer in emerging economies such as China. The CSMAR dataset shows that only 1.8% of the 3,600 Chinese public firms had a formal CIO position in 2018. Consequently, while many firms do have CIOs, CIO presence and its positive implications should not be taken for granted.

Extant research hints that CIO presence could have a positive impact on AI orientation. For example, Carter et al. (2011) suggest that CIO presence likely enhances managerial opportunity recognition and focuses attention on the firm's strategic

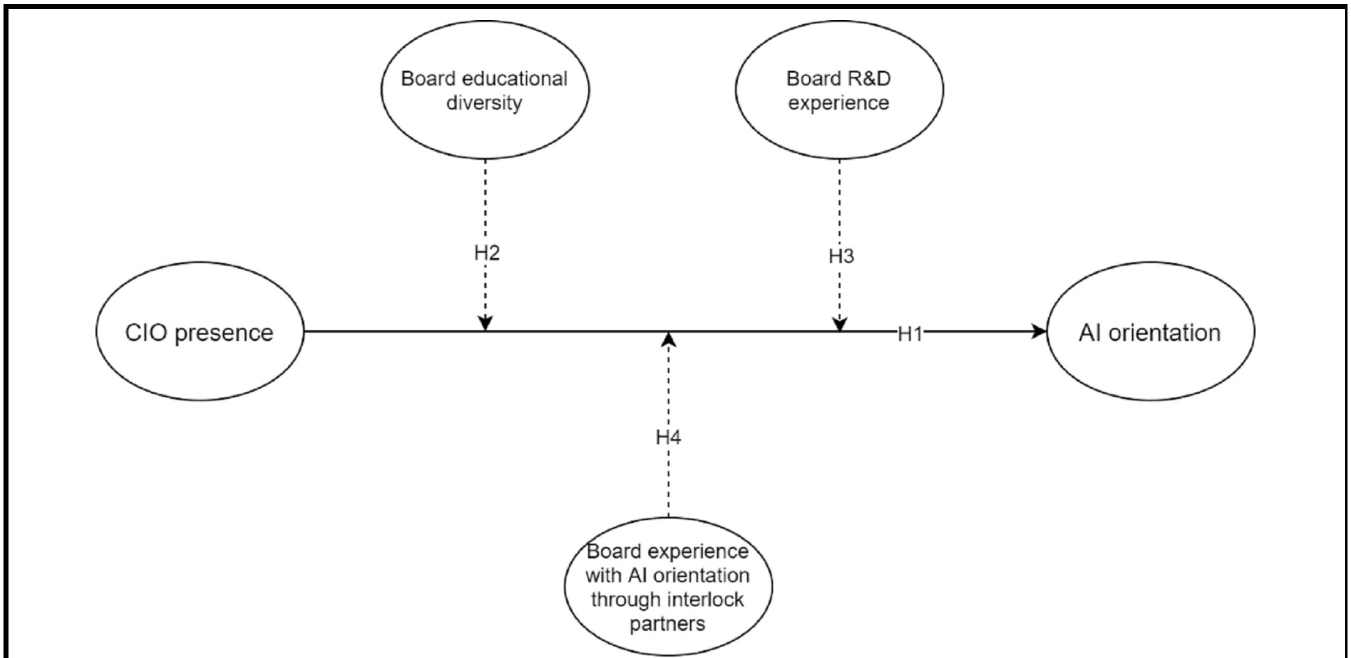


Figure 1. Research Model

application of innovative technologies. If this is true, then CIO presence should not only signal whether a firm prioritizes technological innovations such as AI but should also indicate a firm's commitment to undertaking the risks associated with implementing advanced technologies (Preston et al. 2006). Hence, we explore whether the mere presence of a CIO in the top management team helps predict whether a firm develops an AI orientation.

Our research model is presented in Figure 1, and we discuss our hypotheses below.

CIO Presence and AI Orientation

Existing research on upper echelons theory offers three reasons explaining how the presence of a CIO shapes AI orientation: the CIO's strategic role, technological role, and power. *Strategically*, upper echelons theory argues that the presence of a CIO adds the technological perspective necessary to help members of the upper echelon better understand AI and its potential contribution to improving the firm's competitive advantage (Fu et al. 2020). Specifically, CIOs presumably possess the ability to translate the technical language of AI into operational language that highlights the value of AI orientation for the firm (Kolbjornsrud et al. 2017). For example, based on technology and firm-specific knowledge, a CIO should be able to explain how AI can increase the accuracy of decisions to peers who may lack an understanding of the black box algorithms that shape AI recommendations. Similarly, CIOs should be able to explain why applying machine learning to improve a process requires building an infrastructure capable of acquiring the data needed to continually improve the AI algorithms. By communicating the strategic impact of AI in order to justify associated IT investments, CIOs can motivate upper echelons to support AI orientation.

Technologically, as members of upper echelons, CIOs act as technologists that handle specific IT implementations and as digital strategists that position IT at the abstract strategic level (Carter et al. 2011). As technologists, CIOs assist in evaluating, training, and selecting AI-oriented employees, and work with other IT experts to prepare the overall data environment through, for example, developing an AI-friendly data environment and internal algorithm standards for AI. As digital strategists, CIOs evaluate the possibility and necessity of establishing an AI-friendly organizational environment. Also, CIOs shape understanding of whether and when to make the personnel changes necessary to sustain a workforce capable of working with AI. Acting as either technologists or strategists, CIOs are well-positioned to make the case to upper echelons about technology feasibility and the strategic need to develop an AI orientation.

In terms of *power*, the presence of a CIO in the top management team can facilitate and sustain an AI orientation. Effective construction of a new data environment requires expertise at the top levels of the organization. In the short term, having a CIO invests a member of the upper echelons with the power necessary to gather the required resources and manage the implementation process (Garms and Engelen 2019; Li, Su, and Liu 2010). In the long term, the presence of a CIO can focus power in a role that is essential for translating the technological requirements of AI into strategic action. Furthermore, the CIO has the capacity to motivate other members of the upper echelon to accept and promote AI orientation. In sum, we hypothesize:

> **H1**: The presence of a CIO positively impacts AI orientation.

The Moderating Role of Boards in Upper Echelons Theory

While research based on classical upper echelons theory has documented how top managers can affect organizational outcomes, it suffers from certain limitations. In recent updates of upper echelons theory, Hambrick and colleagues suggest that top managers may lack decision rights because of institutional pressures such as strict regulations, strong norms and common values, or organizational governance structures such as well-established routines and powerful boards. In particular, strategic management scholars direct attention to boards as a mechanism that limits managerial discretion and affects the impact of upper echelons on organizational outcomes (Garcia-Sanchez et al. 2019). While theory advocates for this view, little empirical evidence supports this position. We thus explore the contingency effects of board characteristics by examining whether the interplay between the CIO and the board of directors shapes the development of AI orientation.

In the IS context, boards can play an important role in shaping or approving IT strategy and investments. Boards are responsible not only for helping upper echelons craft business and IT strategies but also for monitoring the progress of an organization toward realizing business and IT strategies (Turedi 2020). Board support is important for CIOs, according to Hambrick, because it mitigates possible resistance to strategy shifts and the investment of resources necessary to effect strategic changes such as the development of an AI orientation (Hambrick et al. 2005). Thus, CIOs will be more likely to win support for AI initiatives if they receive board support for AI orientation and related IT investments (Enns et al. 2007).

In addition to the board's monitoring and strategic responsibilities, board members' interest in AI orientation may result from AI's potential to significantly change their relationships with the upper echelons and other board members. AI can substantially change the role of boards as providers of information resources by easing access to information and by generating accurate and transparent strategic recommendations. Further, to monitor the deployment of AI, boards need technological expertise to understand how AI works and how it will impact the organization. Because they are charged with anticipating how AI will impact the organization, we would expect boards to be motivated to actively engage with CIOs seeking to develop an AI orientation (Cremers et al. 2017).

While our logic suggests that AI orientation is likely to attract board attention, upper echelons theory suggests that the quality of that attention may vary according to board member characteristics and board members' relationships with partners in the external environment. For example, upper echelons theory suggests that board composition can affect how members communicate with the CIO, understand the possibilities presented by AI, and view the resources required to support AI (Boivie et al. 2015; Festinger 1954; Fiegenbaum et al. 1996; Poarc et al. 1999; Sharapov and Ross 2019; Shi et al. 2017). To understand the quality of board attention, we evaluate how board educational diversity, R&D experience, and AI orientation experience through interlock partners impacts the relationship between CIO presence and AI orientation.

Board Educational Diversity

We predict that board educational diversity positively impacts the relationship between CIO presence and AI orientation. We base educational diversity on the tertiary educational experience (e.g., university majors) of board members. Upper echelons research suggests that group educational heterogeneity can affect strategic choices by introducing knowledge and perspective heterogeneity in a way that improves firm innovativeness (Hambrick and Mason 1984; Qian et al. 2013; Talke et al. 2010). Such diversity thus impacts a firm's strategic choices regarding AI orientation by buffering the effects of uncertainty or insufficient experience on the development of an AI orientation.

Higher levels of board educational diversity likely make a firm more apt to develop an AI orientation. Diverse educational backgrounds endow board members with different thoughts, perspectives, and knowledge, and can help prevent self-serving behavior and the phenomenon of group think while fostering creative thinking (Talke et al. 2010). Educationally diverse boards possess different perspectives and a heterogeneous knowledge base and are thus more likely to understand the potential benefits associated with developing an AI orientation to manage the change, risks, and costs associated with AI deployment within the firm.

Boards with a heterogeneous knowledge base and a creative mindset are likely better positioned to understand the CIO's perspective regarding technology implications. Further, such boards are likely to include members with technological expertise, who can help the CIO communicate the capabilities of complex technologies such as AI, clarify how AI maps to firm strategy, and share their understanding and perspectives with less technologically adept board members in formal and informal meetings, mitigating their concerns about AI. Thus, we hypothesize that

> **H2**: Board educational diversity enhances the relationship between CIO presence and AI orientation.

Board R&D Experience

Board R&D experience also supports the relationship between CIO presence and AI orientation. This argument is consistent with work indicating that relevant experience predicts board members' engagement with strategic decision-making (Kroll et al. 2008). We define board R&D experience as the number of board members with R&D-related work experience. Although board R&D experience may or may not be technology related (e.g., industrial design), R&D work generally utilizes innovative methods (or innovative problem solving) to enhance business performance (Lofsten 2014) and is thus relevant to using AI to enhance firm business performance. Thus, boards with high level of R&D experience are better positioned to understand the benefits of supporting AI orientation, evaluate strategic issues associated with AI, tolerate the potential limitations of AI, and advise upper echelons in the development of AI-related policies. Because boards with ample R&D experience are likely to understand the CIO's interest in highly innovative technologies like AI, they can facilitate communication among board members, upper echelons, and the CIO. We also expect boards with high levels of R&D experience to respond positively to the CIO's entrepreneurial activities and support AI orientation development (Pan et al. 2019). In sum, board members with relevant R&D experience will be able to leverage their experience to offer useful suggestions to the CIO and will be better able to understand AI orientation. Therefore, we hypothesize:

> **H3**: Board R&D experience enhances the relationship between CIO presence and AI orientation.

Board Experience with AI Orientation

While upper echelons theory documents that the perceptions and relationships in top management teams affect decision-making and strategic selection (Hambrick and Mason 1984), board capital, especially the relevant experience of the board, can also effectively impact how board members view opportunities (Boivie et al. 2015; Zhu and Westphal 2014). Specifically, previous board member exposure to AI orientation while serving on interlock boards can provide salient examples and narratives about how things should be done (Boivie et al. 2015). This exposure effect is especially strong in the context of AI orientation because board members likely have little access to other strong objective standards related to AI orientation decisions. Having board members with AI orientation experience from serving on other boards, should thus make the board more receptive to CIO suggestions regarding AI orientation. For example, if board members experienced AI orientations on other boards and have already learned how to construct an internal AI-friendly environment, deal with potential AI failures, and prepare for potential boardroom power-dynamic changes, they are apt to be less concerned about the overall AI orientation process. Thus, such boards would be better positioned to help CIOs promote an AI orientation. Therefore, we hypothesize:

> **H4**: Board experience with AI orientations through interlock partners enhances the relationship between CIO presence and AI orientation.

Research Method

Data Sources

To test our hypotheses, we collected objective firm-level data documenting CIO presence and AI orientation from the CSMAR database and from the annual reports of public firms in China. We further validated our data by leveraging other third-party data sources (e.g., the TianYanCha website[2]). In particular, we focused on public Chinese firms operating in high-tech sectors of the manufacturing and service industries, such as telecommunications equipment, automobiles, electronic devices, software services, and information services because these firms typically emphasize the importance of employing AI in operations and innovation.

[2]https://www.tianyancha.com/

To build our dataset, we coded AI orientation from 2016 to 2018, the time frame in which AI-related technologies first attracted extensive public attention. We then collected data from 2011 to 2015 from the CSMAR database to document CIO presence. After removing missing data and merging data sources, our final sample consisted of 1,454 firms. Our primary data structure is cross-sectional because the establishment of CIO positions and AI orientation often spans multiple years.

Variables and Measures

AI orientation: AI orientation is viewed as an overall strategic direction and aspiration of employing AI technology within an organization. We coded this construct based on firms' annual reports, which typically include information about a firm's strategic directions, goals, and initiatives and thus represent an important and commonly used data source in strategic management research. To do this, we applied text-mining techniques (Bao and Datta 2014; Guerreiro et al. 2016) and developed a program to code the firm's annual reports (Janasik et al. 2009; Weiss et al. 2010).

Specifically, we identified seven categories of keywords to trace AI orientation: (1) *machine learning*: deep learning, unsupervised, supervised; (2) *natural language processing* (NLP): text generation, question answering, context extraction, classification, machine translation; (3) *expert systems*; (4) *speech:* speech to text, text to speech; (5) *vision*: image recognition, machine vision; (6) *planning*; and (7) *high-intelligence robotics* (Press 2017). We developed a Python-based program, based on the PyCharm platform (Islam 2015) to identify and extract all keywords from our targeted firms' annual reports from 2016 to 2018 (Witten and Frank 2002; Zafarani et al. 2014). We pretested our Python code on a small sample by manually checking its reliability and then used the Python-based program to create a file that included firm code, year, and the citations of extracted terms.

To analyze the extracted sentences and contents, we interviewed two faculty members specializing in AI at a top university in China. Based on what we learned, four research assistants with master's degrees were trained to analyze the prepared texts (Jackson 2019). To train our research assistants, we provided 30 randomly selected items to each research assistant and asked each of them to complete their coding independently. Their results were then compared, and any disagreements were resolved through discussion and consultation with AI experts. Then, each research assistant independently coded the assigned items and reciprocally confirmed the coding results. Finally, to further avoid potential coding bias, we followed the operationalization presented in prior studies to code AI orientation as a binary variable in order to represent whether a firm had developed an AI orientation (Banker et al. 2011). This variable was coded as 1 if the firm had developed an AI orientation, and 0 otherwise. To examine intercoder agreement, we used the Fleiss kappa value. The test result revealed a κ value of 0.65, suggesting a substantial level of intercoder agreement (Landis and Koch 1977).

CIO presence: We obtained CIO information from the CSMAR database. Consistent with prior research (e.g., Fu et al. 2020), we identified upper echelon job titles containing the phrase "CIO," "chief information officer," or "chief information executive" as CIOs. Then, we manually scrutinized the role descriptions of the managers to ensure the validity of the identification approach. To maximize the validity and reliability of our data, we verified the CIO data with data obtained from firms' official websites and the TianYanCha website. The TianYanCha website is a privately owned platform registered with China's State Administration for Industry and Commerce and relevant regulators. It specializes in big data services and analysis and contains rich information about Chinese companies. After validating CIO presence, we operationalized it as a binary variable in our dataset. Specifically, CIO presence was coded as 1 if the CIO position existed between 2011 and 2015, and 0 otherwise.

Board educational diversity: We obtained board educational information from CSMAR. We coded board educational diversity according to university study majors (Hutzschenreuter and Horstkotte 2013; Schubert and Tavassoli 2020) based on the Directory of Colleges and Universities[3] published by the Ministry of Education of China in 2012. The directory groups university majors into 12 disciplines: philosophy (01), economics (02), law (03), education (04), literature (05), history (06), science (07), engineering (08), agriculture (09), medicine (10), management (12), and art (13). For individuals with multiple majors, we chose the one most closely related to their current job. We measured educational background as a categorical variable and then employed the Blau Index to measure educational diversity. The Blau Index is calculated using the following Equation (1.1):

$$\text{Board Educational Diversity} = 1 - \sum_{j=1}^{j} Boardedu_{i,j}^2 \quad (1.1)$$

where $Boardedu_{i,j}$ is the share of board members with educational background j of the total board members in firm i. A high score on this index indicates variability in the educa-

[3]http://www.cdgdc.edu.cn/xwyyjsjyxx/xwsytjxx/274346.shtml

tional background of board members, while a low score represents greater educational homogeneity.

Board R&D experience: Board members' R&D experience was also obtained from the CSMAR database. We operationalized board R&D experience by counting the total number of board members who had R&D-related experiences (Solal and Snellman 2019). For example, if there were three board members involved in R&D activities for a firm within a specific period, then the value of R&D experience was coded as 3.

Board experience with AI orientation through interlock partners: We measured board experience with AI orientation through interlock partners by examining interlock partners' AI orientations. We operationalized the interlock partners' AI orientation as a binary variable to proxy whether any interlock partner had pursued an AI orientation. This variable was coded as 1 if one of the firm's interlock partners had pursued an AI orientation, meaning that at least one board member had experienced an AI orientation in an interlock company, and 0 otherwise. To identify the firm's interlock partners, we constructed an interlock network on a firm-year basis derived from CSMAR data from 2016 to 2018. An interlock tie exists when two firms share at least one director.

Control variables: We controlled for firm-level, board-level, and industry-level characteristics. Firm age and size affect a firm's ability to undertake innovation activities (Rothaermel and Deeds 2004). We thus controlled for *firm age,* measured as the number of years that the firm had been in operation, and *firm size,* measured as the natural log of the total number of employees. To capture the effect of ownership structure on innovation, we also controlled for *ownership concentration*, which was operationalized as the percentage of the firm's shares owned by its largest shareholder (Zhang et al. 2018). To control for the effect of financial performance, we included *net profit* as a control variable (Bos et al. 2017). *Leverage,* measured as the ratio of long-term debt to the book value of assets, was included to account for firms' cash holdings and capital structure (Swift 2016). The rationale for including this control variable is that firms with less leverage may have more financial resources to invest in new technologies (Luger et al. 2018). Since innovation is costly, we controlled for the natural log of the *total production cost* and *overhead cost* (Baumers et al. 2016). *R&D employee ratio* was derived from the ratio of the number of R&D employees divided by the total number of employees, and *R&D intensity* was operationalized as R&D expenses divided by sales, set as the industrial average if R&D expenses were unavailable.

Board and top executive characteristics may have an impact on AI orientation (Kor 2007). We thus controlled for *board size, top management team (TMT) size, average board age, board overseas, board ownership*, as well as *chair political ties* and *CIO ownership*. We operationalized board size as the total number of board members and TMT size as the number of top managers in a firm. We operationalized average board age as the average age of all board members, and board overseas as the number of board members who had studied or worked overseas. We based board ownership on the number of shares owned by all board members (log-transformed, +1). Following prior research on boards' political ties, we created a dummy variable of chair political ties if the board chair acted as or was a member of the central- or local-level government, People's Congress, or the People's Political Consultative Conference (Chizma et al. 2015). We computed the log-transformed number of shares owned by CIOs to measure CIO ownership.

To control for industrial effects, we considered *industrial competition, primary competitor's AI orientation*, and *industry fixed effects*. We operationalized industrial competition as the number of firms that operate in the same industry (we used the standard three-digit industrial classification issued by the China Securities Regulatory Commission in 2012 to gauge different industries). In line with research on competitors' reference groups and identification, we selected 20 primary competitors (Duysters et al. 2020). We first ranked all firms operating in the same industry according to their total assets between 2016 and 2018 and then selected the 10 nearest competitors whose total assets were higher than those of the focal firm and another 10 nearest competitors whose total assets were lower than those of the focal firm. After identifying the AI orientation of these competitors, we operationalized the primary competitor's AI orientation as the total number of primary competitors who had pursued AI orientations. We further included an industrial dummy to represent two industry categories: manufacturing and services.

Finally, to control for the effects of conventional IT orientation, we included two dummy variables: *ERP orientation* and *non-ERP IT orientation* in the model, using AI orientation as our dependent variable. We also compiled a glossary of terms related to ERP to code ERP orientation and developed another glossary of terms relating to CRM (client relationship management), SCM (supply chain management), CAM (computer-aided manufacturing), and MIS (management information systems) to code non-ERP IT orientation. We strictly followed the same approach used in our coding of AI orientation to code ERP and non-ERP IT orientation.

Analytical Approach

To account for the dichotomous dependent variable, we employed the Probit model with clustering robust standard errors on industry ID[4] (the Logit model produces similar results). Probit regression is a suitable method for the prediction of binary dependent variables (Tabachnick and Fidell 2014). Before testing the hypotheses, we standardized non-binary moderators.

Analysis Results

Table 1 reports the descriptive statistics and correlations of the variables. The variance inflation factors (VIFs) were less than the cutoff of five for all regression models, suggesting that multicollinearity was not a problem for our data. Figure 2 depicts the number trend of the firms that had AI orientations and the seven aspects of AI orientation from 2016 to 2018. All AI terms were increasingly mentioned in the annual reports from 2016 to 2018.

The results of hypothesis testing are represented in Table 2. Model 1 is the baseline model with control variables. Model 2 tests H1; Model 3 tests H2; Model 4 tests H3; Model 5 tests H4, and Model 6 is the full model with all moderators. The significance level of the Wald Chi-squared statistics in all of the models indicates that the explanatory variables explained a significant portion of the variation in the dependent variable.

H1 predicts that the presence of a CIO is positively related to AI orientation. As shown in Model 2, there is a positive and significant relationship between CIO presence and AI orientation ($\beta = 0.652$, $p < 0.01$), which supports H1. H2 and H3 predict, respectively, that board educational diversity and board R&D experience positively moderate the relationship between CIO presence and AI orientation, such that the relationship between CIO presence and AI orientation is more salient when board educational diversity and board R&D experience are respectively high. As shown in Model 3, the interaction between CIO presence and board educational diversity positively affects AI orientation ($\beta = 0.548$, $p < 0.01$), thus supporting H2. Model 4 shows that board R&D experience positively moderates the relationship between CIO presence and AI orientation ($\beta = 0.635$, $p < 0.01$), thus providing strong support for H3. H4 states that board experience with AI orientation through interlock partners enhances the positive relationship between CIO presence and AI orientation. Model 5 suggests that this hypothesis is supported ($\beta = 1.462$, $p < 0.01$). Model 6 further confirms all above hypothesis testing results.

To further verify our hypotheses, we plotted the moderating role of board educational diversity, board R&D experience, and board experience with AI orientation through interlock partners at their respective high and low levels (operationalized at one standard deviation above and below the mean) (Aiken and West 1991). We then conducted a simple slope analysis. The results suggest that the effects of higher values of board educational diversity, board R&D experience, and board experience with AI orientation through interlock partners are more statistically significant than the effects of their respective lower values (see Figure 3).

Robustness Checks

To test the robustness of our results, we conducted a series of additional tests to examine the robustness of our hypothesis testing. We also conducted post hoc tests to evaluate potentially meaningful relationships that were not formally hypothesized but could inform future research. We conducted additional checks described below and brief robustness results are presented in Table 3.

Endogeneity: Given the cross-sectional nature of our dataset, one could plausibly argue that there is a potential bias caused by endogeneity. To account for this possibility, we ran additional models using instrumental variables in place of CIO presence (Bascle 2008; Semykina and Wooldridge 2010). We selected total sales and return on assets (ROA) as instrumental variables because they are indicators of firm financial performance (Rubera and Droge 2013), high levels of sales and ROA can thus also serve as indicators of firm performance. Compared to their counterparts, firms with better financial resources are likely better able to attract more highly skilled jobseekers (Shaw et al. 2013) and should have greater opportunities to equip multiple talents and fill every work position. Similarly, prior studies have suggested that better financial performance may enable firms to utilize advanced IT technologies (e.g., AI-related technologies) for innovation because they have more financial resources to invest (Troilo et al. 2014); however, such firms may also be constrained in this regard because their managers tend to make conservative choices to protect their competitive advantage (Chen et al. 2010). Specifically, total sales and ROA are both related to CIO presence at the 5% significance level (the coefficient is 0.19 for total sales and 0.10 for ROA) but are insignificantly related to AI orientation (the coefficient is 0.04 for total sales and -0.03 for ROA). Using a two-stage least squares (2SLS) model, we found support for the main effect that CIO presence can promote AI orientation ($\beta = 1.416$, $p < 0.05$; see

[4]Industry ID was identified using the 2012 Guideline on Industry Classification of Listed Companies enacted by the China Securities Regulatory Commission.

Table 1. Descriptive Statistics and Correlations

Variables	Mean	STD	1	2	3	4	5	6	7	8	9	10	11
1 AI Orientation	0.18	0.38	1										
2 CIO Presence	0.07	0.25	0.217***	1									
3 Board Educational Diversity	0.97	0.02	0.016	-0.049	1								
4 Board R&D Experience	2.28	1.77	0.068***	-0.002	-0.026	1							
5 Board Experience with AI Orientation through Interlock Partners	0.08	0.68	0.120***	0.049	0.007	-0.008	1						
6 Firm Age	17.52	5.48	-0.004	0.024	0.012	-0.157***	0.046	1					
7 Firm Size	7.7	1.35	0.131***	0.145***	-0.077***	-0.091***	0.088***	0.179***	1				
8 Ownership Concentration	0.32	0.14	-0.093***	-0.039	-0.011	-0.048	-0.051	-0.02	0.138***	1			
9 Net Profit	3.15	10.07	0.021	0.237***	-0.139***	-0.053**	0.054**	0.105***	0.469***	0.086***	1		
10 Leverage	0.04	0.26	-0.032	-0.019	0.011	0.028	0.025	-0.166***	-0.300***	-0.099***	-0.017	1	
11 Total Production Cost	20.6	1.95	0.073***	0.126***	-0.050	-0.134***	0.060**	0.264***	0.816***	0.176***	0.413***	-0.579***	1
12 Overhead Cost	18.17	1.78	0.139***	-0.087***	0.003	0.000	0.021	0.123***	0.543***	0.171***	-0.103***	-0.459***	0.611***
13 R&D Employee Ratio	0.17	0.14	0.170***	0.031	-0.054**	0.310***	-0.008	-0.180***	-0.235***	-0.106***	-0.162***	-0.059**	-0.206***
14 R&D Intensity	0.04	0.05	0.140***	0.006	-0.066**	0.242***	0.014	-0.156***	-0.166***	-0.116***	-0.152***	-0.047	-0.231***
15 Board Size	10.46	2.48	-0.038	0.131***	-0.014	0.159***	0.035	0.090***	0.176***	-0.051	0.321***	0.024	0.154***
16 TMT Size	6.44	2.5	0.085***	0.115***	-0.016	0.049	0.052**	0.032	0.382***	0.008	0.213***	-0.121***	0.359***
17 Average Board Age	51.56	3.39	0.025	0.052**	-0.007	0.055**	0.062**	0.128***	0.282***	0.101***	0.262***	-0.046	0.258***
18 Board Overseas	1.62	1.82	0.031	0.128***	-0.434***	0.107***	0.076***	0.018	0.227***	-0.025	0.361***	-0.016	0.183***
19 Board Ownership	13.81	6.45	0.060**	-0.002	-0.007	0.135***	-0.008	-0.228***	-0.049	-0.216***	-0.105***	-0.139***	-0.042
20 Chair Political Ties	0.04	0.18	0.054**	0.02	-0.016	0.017	0.038	0.004	0.113***	-0.036	0.144***	0.128***	0.024
21 CIO Ownership	0.14	1.35	0.056**	0.394***	-0.070***	0.027	0.031	0.018	0.115***	-0.034	0.173***	-0.001	0.113***
22 Industry Competition	193.9	90.41	0.128***	-0.080***	-0.001	0.191***	0.003	-0.121***	-0.076***	-0.077***	-0.243***	-0.075***	-0.165***
23 Primary Competitor's AI Orientation	3.08	2.86	0.272***	0.090***	-0.021	0.166***	0.093***	-0.091***	0.201***	-0.092***	-0.063**	-0.019	0.096***
24 Industry Dummy	0.73	0.44	-0.038	-0.181***	0.060**	0.151***	0.017	-0.090***	-0.016	0.062**	-0.140***	-0.012	-0.105***
25 ERP Orientation	0.14	0.35	0.386***	0.171***	0.02	0.006	0.094***	0.037	0.084***	-0.035	-0.019	-0.022	0.062**
26 NonERP Orientation	0.07	0.26	0.312***	0.128***	-0.004	-0.003	0.046	-0.018	0.082***	-0.039	0.005	-0.025	0.069***

Variables	12	13	14	15	16	17	18	19	20	21	22	23	24	25
12 Overhead Cost	1													
13 R&D Employee Ratio	0.073***	1												
14 R&D Intensity	0.076***	0.608***	1											
15 Board Size	-0.203***	-0.147***	-0.128***	1										
16 TMT Size	0.261***	0.01	0.037	0.147***	1									
17 Average Board Age	0.132***	-0.084***	-0.083***	-0.015	0.163***	1								
18 Board Overseas	-0.017	0.006	0.039	0.246***	0.162***	0.091***	1							
19 Board Ownership	0.053**	0.229***	0.204***	-0.114***	0.058**	-0.154***	0.066**	1						
20 Chair Political Ties	-0.001	-0.053**	-0.035	0.065**	0.097***	0.080***	0.084***	0.016	1					
21 CIO Ownership	-0.115***	-0.008	-0.013	0.130***	0.161***	0.064**	0.115***	0.033	0.078***	1				
22 Industry Competition	0.133***	0.307***	0.317***	-0.236***	-0.071***	-0.081***	0.013	0.180***	-0.055**	-0.082***	1			
23 Primary Competitor's AI Orientation	0.220***	0.344***	0.266***	-0.078***	0.107***	0.002	0.059**	0.149***	-0.002	0.015	0.304***	1		
24 Industry Dummy	0.119***	-0.064**	0.038	-0.166***	0.000	0.047	-0.024	0.046	-0.021	-0.089***	0.533***	-0.053**	1	
25 ERP Orientation	0.103***	0.006	0.021	-0.048	0.043	0.029	0.003	0.019	0.013	0.008	0.092***	0.107***	-0.009	1
26 NonERP Orientation	0.097***	0.064**	0.022	-0.015	0.088***	-0.003	0.012	0.036	-0.008	-0.006	0.012	0.088***	-0.089***	0.382***

Notes: **p < 0.05, ***p < 0.01.

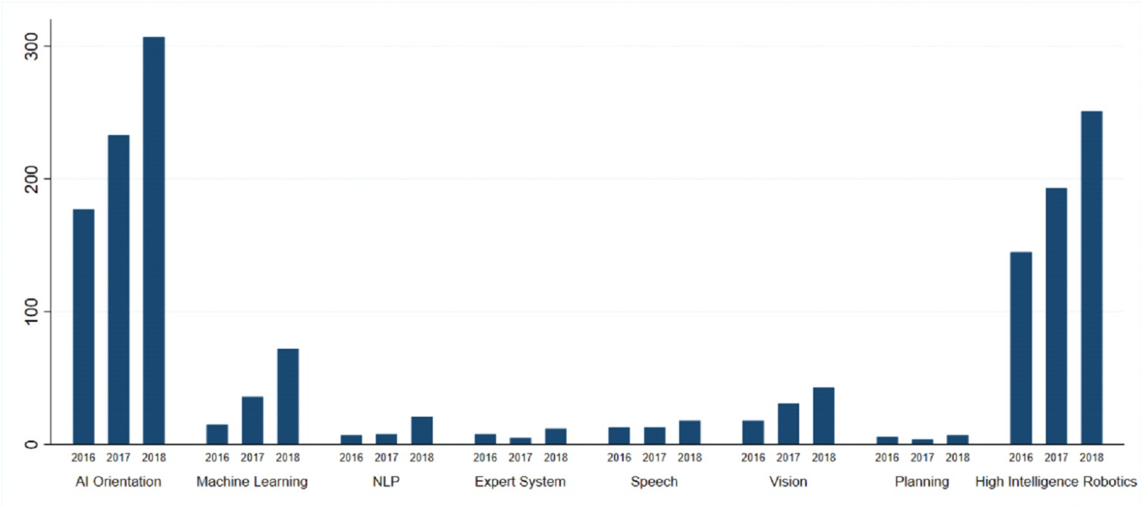


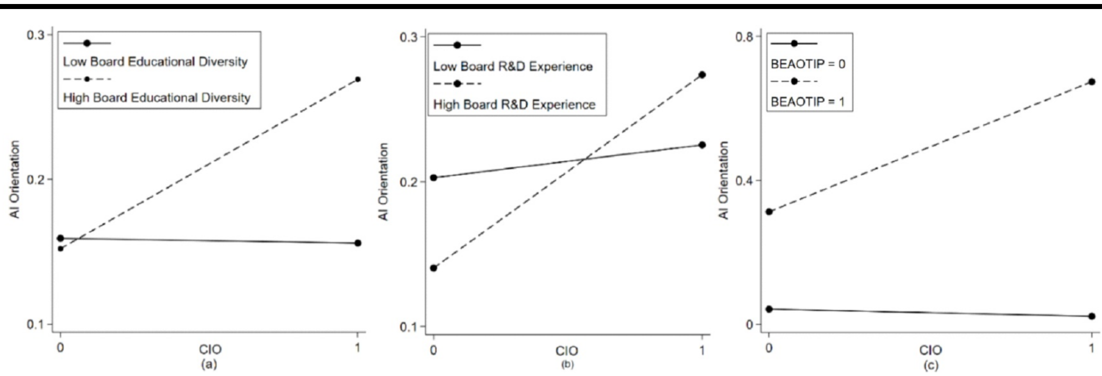
Figure 2. The Number Trend of Firms Orienting AI

Table 2. Regression Results

Variables	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Constant	-3.696*** (-3.63)	-3.853*** (-3.75)	-3.692*** (-3.90)	-4.111*** (-3.98)	-3.824*** (-3.64)	-3.905*** (-4.04)
Firm Age	0.004 (0.52)	0.004 (0.48)	0.004 (0.52)	0.005 (0.55)	0.005 (0.62)	0.006 (0.69)
Firm Size	0.067 (0.98)	0.063 (0.94)	0.074 (1.10)	0.059 (0.92)	0.056 (0.85)	0.063 (0.97)
Ownership Concentration	-0.725*** (-2.77)	-0.730*** (-2.68)	-0.683*** (-2.64)	-0.691** (-2.46)	-0.713*** (-2.64)	-0.639** (-2.40)
Net profit	0.011** (2.16)	0.009 (1.83)	0.009** (1.98)	0.008 (1.66)	0.010** (2.26)	0.010** (2.14)
Leverage	-1.145 (-1.06)	-1.020 (-0.95)	-1.135 (-1.02)	-1.039 (-0.96)	-0.899 (-0.87)	-1.015 (-0.95)
Total Production Cost	-0.042 (-0.56)	-0.050 (-0.65)	-0.041 (-0.54)	-0.055 (-0.69)	-0.048 (-0.60)	-0.049 (-0.59)
Overhead Cost	0.122** (1.98)	0.140** (2.15)	0.117** (2.11)	0.153** (2.15)	0.141** (2.12)	0.136** (1.97)
R&D Employee Ratio	0.807** (2.15)	0.786** (2.04)	0.845** (2.21)	0.776** (2.02)	0.850** (2.14)	0.893** (2.19)
R&D Intensity	0.671 (1.19)	0.647 (1.23)	0.638 (1.28)	0.607 (1.11)	0.649 (1.16)	0.575 (1.07)
Board Size	0.012 (0.58)	0.010 (0.48)	0.005 (0.25)	0.014 (0.66)	0.009 (0.48)	0.009 (0.44)
TMT Size	-0.006 (-0.29)	-0.009 (-0.45)	-0.011 (-0.56)	-0.009 (-0.44)	-0.012 (-0.60)	-0.013 (-0.66)
Average Board Age	0.002 (0.11)	0.002 (0.12)	0.003 (0.13)	0.004 (0.18)	0.002 (0.10)	0.003 (0.17)
Board Overseas	0.032 (1.12)	0.032 (0.98)	0.040 (1.49)	0.034 (1.11)	0.035 (1.09)	0.045 (1.73)

Variables	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Chair Political Ties	-0.012** (-2.43)	-0.011** (-2.27)	-0.012** (-2.40)	-0.011** (-2.08)	-0.011** (-2.14)	-0.011** (-2.04)
Board Ownership	-0.117 (-0.40)	-0.088 (-0.28)	-0.078 (-0.25)	-0.094 (-0.29)	-0.111 (-0.32)	-0.112 (-0.32)
CIO Ownership	0.071** (2.19)	0.028 (1.04)	0.046 (1.53)	0.010 (0.31)	0.034 (1.14)	0.029 (0.84)
Industry Competition	0.001 (0.84)	0.000 (0.69)	0.000 (0.53)	0.001 (0.88)	0.000 (0.52)	0.000 (0.53)
Primary Competitor's AI Orientation	0.083*** (2.69)	0.081*** (2.62)	0.083*** (2.67)	0.078** (2.48)	0.080*** (2.72)	0.082*** (2.65)
Industry Dummy	-0.047 (-0.26)	0.019 (0.10)	0.031 (0.16)	0.004 (0.02)	0.010 (0.06)	0.009 (0.05)
ERP Orientation	1.004*** (7.15)	0.953*** (7.54)	0.963*** (7.77)	0.962*** (7.68)	0.946*** (7.70)	0.971*** (8.00)
NonERP Orientation	0.772*** (6.12)	0.735*** (5.69)	0.789*** (6.04)	0.739*** (5.73)	0.764*** (6.05)	0.803*** (6.35)
Board Educational Diversity	0.120 (1.43)	0.121 (1.30)	0.080** (2.06)	0.113 (1.32)	0.111 (1.26)	0.082* (1.96)
Board R&D Experience	-0.056 (-1.06)	-0.044 (-0.77)	-0.045 (-0.77)	-0.085 (-1.49)	-0.042 (-0.72)	-0.080 (-1.35)
Board Experience with AI Orientation through Interlock Partners (BEAOTIP)	1.114*** (8.13)	1.107*** (8.08)	1.104*** (7.95)	1.113*** (8.03)	1.061*** (7.42)	1.070*** (7.39)
CIO Presence		0.652*** (3.05)	0.681*** (3.72)	0.861*** (3.25)	0.706*** (3.17)	0.956*** (4.05)
CIO Presence × Board Educational Diversity			0.548*** (3.05)			0.524*** (3.24)
CIO Presence × Board R&D Experience				0.635*** (2.94)		0.635** (2.13)
CIO Presence × BEAOTIP					1.462*** (3.72)	1.382*** (2.97)
Observations	1454	1454	1454	1454	1454	1454
Pseudo R²	0.411	0.42	0.425	0.425	0.426	0.433

Notes: Robust z-statistics in parentheses. **p < 0.05, ***p < 0.01.



In the legend of Figure 3(c), we use the abbreviation BEAOTIP to represent board experience with AI orientation through interlock partners.

Figure 3. The Moderating Effect

Table 3. Robust Results

Variables	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9	Model 10	Model 11	Model 12
Board Educational Diversity	0.282** (2.11)	-0.093 (-0.23)	0.080** (2.06)	-0.017 (-0.75)	-0.021 (-0.85)	0.943*** (3.73)	0.633 (0.74)	0.018 (0.27)	-0.002 (-0.03)	0.145** (2.17)	0.120*** (2.93)	0.012 (0.20)
Board R&D Experience	0.005 (0.85)	0.090** (2.49)	-0.045 (-0.77)	-0.007 (-0.28)	-0.014 (-0.49)	1.423 (2.06)	0.850 (0.93)	-0.050 (-1.02)	-0.030 (-0.63)	-0.044 (-1.52)	-0.044 (-1.43)	-0.077 (-1.42)
Board Experience with AI Orientation through Interlock Partners (BEAOTIP)	0.079*** (2.83)	4.437*** (9.87)	1.104*** (7.95)	0.020 (1.05)	0.018 (0.98)	-1.920 (-1.69)	-2.597 (-1.52)	1.109*** (7.78)	1.059*** (7.17)	1.101*** (5.98)	1.152*** (6.48)	1.055*** (7.03)
CIO Presence	1.416** (2.19)	0.682*** (2.75)	0.681*** (3.72)	0.238** (2.21)	0.257** (2.09)	4.809** (2.35)	15.111** (2.06)	0.266** (2.00)	0.346** (2.06)	0.126*** (3.13)	0.173*** (4.69)	0.976*** (3.99)
CIO Presence × Board Educational Diversity			0.497** (2.31)		0.095** (2.04)		5.428 (1.56)		0.170** (1.99)		0.075** (1.99)	0.528*** (2.66)
CIO Presence × Board R&D Experience			-0.495** (-2.52)		0.140* (1.69)		16.327** (3.53)		-0.129 (-0.53)		0.118** (2.11)	0.697** (2.32)
CIO Presence × BAOEIIP			15.455*** (3.51)		0.800** (2.43)		18.250** (2.07)		1.447*** (3.83)		0.397*** (2.83)	1.590*** (2.97)

Variables	Model 13	Model 14	Model 15	Model 16	Model 17	Model 18	Model 19	Model 20
Board Educational Diversity	0.082 (1.92)	0.049 (1.31)	0.218 (1.39)	0.043 (0.27)	0.118 (1.20)	0.081 (1.87)	0.116*** (2.61)	0.098*** (3.17)
Board R&D Experience	-0.079 (-1.05)	0.006 (0.12)	0.314** (2.37)	0.227 (1.24)	-0.045 (-0.79)	-0.077 (-1.31)	0.049 (0.90)	0.047 (0.84)
Board Experience with AI Orientation through Interlock Partners (BEAOTIP)	1.071*** (7.56)	-0.502 (-0.82)	1.192*** (5.79)	1.092*** (5.73)	1.112*** (7.97)	1.073*** (7.40)	1.164*** (7.20)	1.152*** (7.23)
CIO Presence	0.251 (0.89)	1.062*** (6.50)	0.681** (2.44)	0.974** (1.98)	0.660*** (3.26)	0.965*** (4.41)	0.708** (2.12)	0.985** (2.14)
CIO Presence × Board Educational Diversity	0.082*** (3.74)	0.554*** (3.41)		0.582** (2.12)		0.514*** (3.12)		0.373** (2.07)
CIO Presence × Board R&D Experience	2.616** (2.01)	0.648*** (2.86)		0.496 (0.97)		0.631** (2.17)		0.041 (0.10)
CIO Presence × BEAOTIP	1.346*** (2.99)	0.866** (2.51)		1.015** (2.35)		1.401*** (2.91)		2.883*** (3.01)

Notes: Robust z-statistics in parentheses; **p < 0.05, ***p < 0.01. Same control variables are applied to each model.

Model 1 in Table 3). Several post-estimation tests were used to validate total sales and ROA. A test of overidentification was conducted to assess the exogeneity of our instruments; the Wooldridge's robust score was insignificant (score $\chi^2 = 1.058$, $p = 0.303$), suggesting that our instruments are valid and/or that our model is correctly specified (Bascle 2008). We implemented a test of weak instruments and found the *F*-statistic to be 10.35. According to Stock et al. (2002), this test indicates that our instruments are strong since the *F*-statistic is greater than 10.

Further, we considered an alternative analysis with panel data to mitigate time-related endogeneity. We obtained our data from the same sources used for the firm-year basis; however, in the panel dataset, we employed a three-year running AI orientation to capture the extent to which firms pursued AI orientations from 2009 to 2018. Using this panel sample, we retested our proposed hypotheses and found support for our main effect ($\beta = 0.682$, $p < 0.01$) (see Model 2 in Table 3), the moderation of board educational diversity ($\beta = 0.497$, $p < 0.05$), and board experience with AI orientation through interlock partners ($\beta = 15.455$, $p < 0.01$) (see Model 3 in Table 3).

Alternative operationalization of AI orientation: Since researchers have found that media coverage is a useful way to study firms' strategic direction (Zhang 2006), we used data from media coverage of firms' AI orientations as an alternative measure of AI orientation—specifically, media coverage from 2016 to 2018 relating to our sample companies from the Financial News Database of Chinese Listed Companies (CFND)[5] on the Chinese Research Data Services Platform. We adopted the same coding process used for the original AI orientation to code the news reports and generate the binary variable of media coverage of AI orientation. We reran our models with the same model specifications and control variables used previously and found support for our main effect ($\beta = 0.238$, $p < 0.05$) (see Model 4 in Table 3) as well as the moderation of board educational diversity ($\beta = 0.095$, $p < 0.05$), board R&D experience ($\beta = 0.140$, $p < 0.1$), and board experience with AI orientation through interlock partners ($\beta = 0.800$, $p < 0.05$) (see Model 5 in Table 3). When measuring AI orientation as the amount of total media coverage using the keyword "AI orientation," we found support for our main effect ($\beta = 4.809$, $p < 0.05$) (see Model 6 in Table 3) as well as the moderation of board R&D experience ($\beta = 16.327$, $p < 0.05$) and board experience with AI orientation through interlock partners ($\beta = 18.250$, $p < 0.05$) (see Model 7 in Table 3).

As Figure 2 suggests, AI orientation comprises seven categories, with high-intelligence robotics being the largest. We thus performed a sensitivity test on those seven categories, using three alternatives to measure AI orientation: the first included high-intelligence robotics, the second included the other six aspects, and the third included a randomly selected combination of high-intelligence robotics and any one of the other six categories. We then tested our hypotheses using these three alternatives. We replicated our results (shown in Table 2) when adopting the first and the third alternatives, respectively. Using the second alternative, we found support for our main effect of CIO presence as well as the moderation of board educational diversity and board experience with AI orientation through interlock partners.

Alternative operationalization of independent variables: We further checked the robustness of our model by using alternative measures of CIO presence in both the cross-sectional sample and the panel sample, and our results remained largely the same. First, since CTOs may assume the role of CIOs in certain firms, we added firms with CTO positions to our sample and retested our model, finding support for our main effect ($\beta = 0.266$, $p < 0.05$) (see Model 8 in Table 3) as well as the moderation of board educational diversity ($\beta = 0.170$, $p < 0.05$) and board experience with AI orientation through interlock partners ($\beta = 1.447$, $p < 0.01$) (see Model 9 in Table 3). Second, we considered the ordinal measure of CIO presence based on the CIO's structural position in top management teams. We first operationalized CIO structural power as a categorical variable: CIO structural power was coded as 1 if there was a CIO present, 2 if the CIO was part of the TMT, 3 if CIO sat on the board, and 0 if no CIO was present.[6] When introducing these new CIO structural power measures with the same model specifications and all other variables unchanged, we were able to replicate our results, which are presented in Table 3 (see Models 10 and 11 in Table 3).

Third, we replaced board educational diversity with board professional background diversity, which can also reflect board members' overall perspectives and knowledge base. Specifically, we coded board professional diversity according to ten categories: production (1), R&D (2), design (3), human resources (4), management (5), marketing (6), finance (7), accounting (8), law (9), other factors (10). Similarly, for board members with multiple professional backgrounds, we chose the one that most closely represented their current job. When adopting this alternative measure to replace educational diversity, we found support for the moderation of board professional background diversity ($\beta = 0.528$, $p < 0.01$), with all other results remaining significant (see Model 12 in Table 3).

We also adopted the alternative measure of board R&D experience by replacing the total number of board members who had R&D experience with the ratio of board members who had R&D experience compared to the total number of board members, finding support for this replacement ($\beta = 2.616$, $p < 0.05$) (see Model 13 in Table 3). All other results remained largely unchanged. Further, we replaced the moderation of board experience with AI orientation through interlock partners with second-order interlock partners' AI orientation because second-order interlock partners may provide some remote reference. We found support for H4 after using this alternative measure ($\beta = 0.866$, $p < 0.05$), with all other results remaining significant (see Model 14 in Table 3).

Alternative analytical approach: As another robustness check, we also leveraged the coarsened exact matching (CEM) algorithm (Iacus et al. 2012) to match firms with CIO presence to a control group based on firm-level characteristics. Of the remaining candidates, we selected the nearest neighbor based on five firm-level characteristics: industry type, ROI (return on investment), R&D intensity, R&D employee, board R&D experience, and board educational level.[7] CEM regularly produced a dataset of 406 firms (66

[5]So far, the CFND contains 60 million news releases from more than 400 main media organizations (https://www.cnrds.com).

[6]As reported in Table 2, results were consistent when using another ordinal measure of CIO presence (CIO presence was coded as 1 if a CIO was present, 2 if a CIO sat on the board, and 0 if a CIO was not present).

[7]Industry type was identified by the Guideline on Industry Classification of Listed Companies enacted by the China Securities Regulatory Commission in 2012; ROI was computed as the ratio of profit before tax to the total investment. R&D intensity was calculated as R&D expenses divided by

CIO firms and 340 matching firms) and the multivariate imbalance value decreased from 0.98 to 0.80 (Iacus et al. 2012). Then, we retested our model and found support for the main effect ($\beta = 0.681$, $p < 0.05$) (see Model 15 in Table 3) as well as the moderation of board educational diversity ($\beta = 0.582$, $p < 0.05$) and board experience with AI orientation through interlock partners ($\beta = 1.015$, $p < 0.05$) (see Model 16 in Table 3).

Moreover, since the firms in our sample may not be independent, thus resulting in firm correlation, we reran the Probit model, clustering robust standard errors on firm ID. The analysis produced the same research findings. We further replaced the 2011–2015 window of CIO presence with a 2013–2015 window (see Models 17 and 18 in Table 3) and a 2016–2018 window (see Models 19 and 20 in Table 3), finding that our results remained largely unchanged. Overall, our reported findings cannot be accounted for by alternative measures and different estimation specifications, reaffirming the robustness of our model specification.

The distinctiveness of AI orientation: We evaluated whether AI orientation differs from conventional IT orientation. Following prior research suggesting that boards pay little attention to conventional IT (Huff et al. 2006), we argue that conventional ITs, including ERPs, are more widely recognized and trusted to provide benefits to firms with relatively few risks and/or uncertainties. We thus expect that boards may more easily reach consensus on the benefits of conventional IT, regardless of the diversity and experience of their members, and may thus fail to actively contribute to the evaluation of conventional IT-related issues. In contrast, we argue that boards will remain active when considering AI orientation (Tuggle et al. 2010). Therefore, board characteristics may exert a much stronger influence on AI orientation than on the conventional IT orientation. To empirically validate the differences, we conducted the following tests.

First, we tested the effect of CIO presence and the three proposed moderations on conventional IT orientation, including ERP orientation, and non-ERP IT orientation, with the same control variables and model specifications. The results are shown in Table 4. The measurements of ERP orientation and non-ERP IT orientation are described in the Control Variables section above. We found that CIO presence promotes ERP orientation but not non-ERP IT orientation. Further, none of the moderation effects of board characteristics remained significant and positive when the dependent variable was ERP orientation or non-ERP IT orientation. These results largely support our argument that boards' influence will be more salient when evaluating AI orientation.

Further, we examined the direct effect of board educational diversity, board R&D experience, and board AI/ERP/non-ERP IT orientation experience through interlock partners on the target firm's AI orientation, ERP orientation, and non-ERP IT orientation. Model 1 and Model 3 in Table 4 report the results. We found that board educational diversity, board R&D experience, and board ERP/non-ERP IT orientation experience through interlock partners failed to significantly promote ERP orientation and non-ERP IT orientation, respectively. These findings support our argument that AI orientations attract more board attention than conventional IT orientations.

Discussion

Motivated by a desire to better understand managerial factors that can help firms develop AI orientations, we used an upper echelons perspective to explain the effect of CIO presence on AI orientation and also evaluated the moderating effects of board characteristics—namely, board educational diversity, board R&D experience, and board experience with AI orientation through interlock partners—on the relationship between CIO presence and AI orientation.

We found that CIO presence and board characteristics have a significant positive impact on AI orientation. While we consistently found that CIO presence impacted AI orientation, ERP orientation, and other more conventional IT orientations, our results suggest that board engagement only impacts AI orientation, a new but distinctive form of IT strategic action. These findings confirm that while CIOs are important for the formation and maintenance of all forms of IT orientation (Chatterjee et al. 2001), board characteristics positively moderate the relationship between CIO presence and the development of AI orientation, suggesting that that the distinctive features of AI either attract board attention or require the CIO to solicit board support for AI orientation (Caluwe and Haes 2019). To the best of our knowledge, our study is the first to examine the interplay between CIOs and boards for formulating IT orientation in general and AI orientation in particular.

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sales and was set to the industrial average if missing. R&D employees is the ratio of the number of R&D employees to the total number of employees. Board R&D experience was calculated as the ratio of R&D board members to the total number of all members of the board. Board education level was based on the number of board members with postgraduate degrees divided by board size.

Table 4. Regression Results Predicting ERP and Non-ERP IT Orientation

Variables	DV = ERP		DV = NON-ERP-IT	
	Model 1	Model 2	Model 3	Model 4
Constant	-3.932*** (-3.40)	-4.401*** (-3.50)	-5.635*** (-2.40)	-6.119*** (-2.79)
Firm Age	0.013 (1.31)	0.013 (1.28)	-0.013 (-1.58)	-0.012 (-1.39)
Firm Size	-0.053 (-0.93)	-0.052 (-0.91)	-0.016 (-0.12)	-0.031 (-0.22)
Ownership Concentration	-0.083 (-0.26)	-0.084 (-0.27)	-0.306 (-0.82)	-0.359 (-0.92)
Net profit	-0.011 (-1.41)	-0.011 (-1.43)	-0.007 (-1.15)	-0.009 (-1.47)
Leverage	0.292 (1.15)	0.289 (1.16)	-3.837** (-2.50)	-3.677** (-2.36)
Total Production Cost	0.029 (0.25)	0.026 (0.23)	0.027 (0.22)	0.012 (0.09)
Overhead Cost	0.113 (1.08)	0.116 (1.08)	0.226*** (2.70)	0.281*** (3.72)
R&D Employee	-1.518** (-2.89)	-1.512** (-2.90)	0.547 (1.81)	0.533 (1.75)
R&D Intensity	0.566 (0.34)	0.568 (0.34)	-1.915 (-1.29)	-1.872 (-1.41)
Board Size	-0.031 (-1.84)	-0.031 (-1.92)	0.006 (0.15)	0.011 (0.29)
TMT Size	-0.009 (-0.48)	-0.009 (-0.49)	0.042 (1.39)	0.043 (1.43)
Average Board Age	0.006 (0.47)	0.006 (0.49)	-0.013 (-0.72)	-0.016 (-0.87)
Board Overseas	0.016 (0.56)	0.015 (0.545)	-0.013 (-0.38)	-0.038 (-1.32)
Chair Political Ties	0.040 (0.14)	0.043 (0.15)	-0.243 (-0.72)	-0.396 (-1.21)
Board Ownership	-0.002 (-0.21)	-0.002 (-0.22)	0.005 (0.56)	0.006 (0.62)
CIO Ownership	-0.037 (-1.11)	-0.035 (-0.95)	-0.085 (-1.56)	-0.200*** (-3.29)
Industry Competition	0.002** (2.10)	0.002** (2.10)	0.000 (0.30)	0.000 (0.17)
Primary Competitor's AI Orientation	-0.008 (-0.26)	-0.008 (-0.25)	-0.032** (-2.10)	-0.045** (-2.50)
Industry Dummy	-0.115 (-1.01)	-0.110 (-1.01)	-0.333*** (-2.65)	-0.337*** (-2.71)
NonERP Orientation	1.204*** (10.13)	1.206*** (9.85)		
AI Orientation	1.002*** (9.09)	1.006*** (9.02)	0.745*** (6.58)	0.806*** (6.86)
ERP Orientation			1.114*** (10.15)	1.099*** (10.04)
Board Educational Diversity	0.059 (0.95)	0.051 (0.83)	-0.021 (-1.09)	0.018 (0.26)

Variables	DV = ERP		DV = NON-ERP-IT	
	Model 1	Model 2	Model 3	Model 4
Board R&D Experience	0.053 (1.03)	0.056 (1.03)	-0.058 (-1.115)	-0.043 (-1.03)
Board ERP Orientation Experience through Interlock Partners (BEOETIP)	-0.016 (-0.23)	-0.034 (-0.47)		
Board NonERP Orientation Experience through Interlock Partners (BNEOETIP)			0.078 (-0.58)	-0.057 (-1.83)
CIO Presence	0.574*** (2.94)	0.440*** (2.59)	0.232 (1.25)	0.305 (1.71)
CIO Presence × Board Educational Diversity		-0.016 (-0.10)		-0.867*** (-3.60)
CIO Presence × Board R&D Experience		-0.046 (-0.28)		0.083 (0.31)
CIO Presence × BEOETIP		0.196 (1.29)		
CIO Presence × BNEOETIP				-0.484 (-1.19)
Observations	1454	1454	1454	1454
Pseudo R^2	0.246	0.247	0.296	0.314

Notes: Robust z-statistics in parentheses; $^{**}p < 0.05$, $^{***}p < 0.01$.

The Relevance of Upper Echelons to Forming an AI Orientation

This paper makes two important theoretical contributions. First, this study contributes to the IT management literature by offering a strategic perspective on how upper echelons influence the development of AI orientation, thus going beyond existing literature, which largely takes a technological perspective to investigate AI (Alsheibani et al. 2019). Extending Enns et al. (2007, p. 29), we provide evidence that CIO presence interacts with boards to influence AI orientation. Specifically, we demonstrate that the interplay of CIO presence with board knowledge heterogeneity (e.g., educational diversity, R&D experience, experience with AI orientation) shapes the formation of AI orientation. By doing so, we advance the understanding of how CIOs and boards impact the strategic management of AI in firms. While acknowledging the potential relevance of general board attributes (e.g., size of board, age of board), organizational characteristics (e.g., organization age, the role of IT in the organization), and contextual factors to IT strategy, Caluwe and Haes (2019) argue that IS research has largely left unaddressed how or why boards engage with major initiatives such as digital transformation. Within the context of AI, we argue that boards engage with IT strategy when changes pose significant cost, risk, or disruptions to existing firm processes and provide evidence of the specific characteristics that affect board attention. Thus, our study lays a foundation for further work on why boards pay attention to AI orientation.

Second, we underscore the uniqueness of AI by demonstrating that the formation of an AI orientation differs from conventional IT orientations. Our post hoc analysis indicates that AI orientations draw board attention whereas conventional IT orientations do not. We believe that AI orientations attract board attention because of the cost associated with deploying AI in an organization. However, the implications of this finding require further investigation. Specifically, it would be useful to further probe drivers of board engagement, such as financial and structural concerns about the demands that AI places on the organization. Future research should also further examine the motivations and effects of board engagement. For example, does the board engage in order to retain power in the organization? Does board engagement resolve agency problems between CIOs and less technologically sophisticated upper echelon members? Or, does board engagement signal a broader agency problem involving technologically oriented boards solidifying control by promoting costly IT investments in order to reify their own power? Therefore, although we found that boards can help promote AI orientation, further research is necessary to clarify why boards engage and to identify the broader implications of board engagement.

Implications for Practice

Our findings underscore the importance of firms formally appointing a CIO as a mechanism to direct the development of an AI orientation. To prepare for the changes that AI is bringing to markets, firms should consider establishing a CIO position and be prepared to address any potential impacts that may affect upper echelon power distribution or reporting structures caused by changes to upper echelon composition. Furthermore, our study suggests that practitioners should pay attention to how board characteristics shape firms' ability to develop an AI orientation. For CIOs interested in formulating an AI orientation, our findings suggest that they should develop strong relationships with board members, particularly those who possess relevant knowledge.

Second, for firms seeking to deploy AI, our work indicates that upper echelons should direct attention to the environment, structure, and culture of the organization, as well as to the composition and power dynamics among upper echelons and boards. In particular, our work indicates the importance of recruiting board members with relevant knowledge and experience (e.g., educational diversity or AI orientation), as well as board members familiar with the challenges of deploying change (e.g., R&D). This is particularly important at the early stages of AI implementation because a firm's upper echelons and employees may not yet possess the knowledge necessary to assess whether the planned AI implementation will be capable of meeting performance goals.

Limitations and Future Research Directions

This study has a number of limitations. First, our work constitutes just one snapshot of strategic AI orientation in firms. Absent internal firm data, it is difficult to capture the full lifecycle of how firms manage AI. Since the inferences from this study are limited by the data and contexts we used, our study will be most relevant to firms seeking to understand drivers of AI orientation at early stages of strategic AI management (Peng et al. 2005). Future studies could monitor the progress of AI deployment in a firm, and longitudinal, qualitative/quantitative studies would offer strong complements to this study.

Second, because AI may have far-reaching implications regarding business organization in firms locally and/or internationally, future studies should explore the effect of CIOs and boards on AI orientation in different cultural and market contexts. We collected data exclusively from public Chinese firms that were active in specific sectors. While this approach increased the internal validity of our study, it may limit the applicability of our findings to companies active in other cultural and economic contexts. This is important because evidence suggests that China may differ not only in terms of values (e.g., time orientation, collectivism, uncertainty avoidance), as compared to the United States (Hofstede and Bond 1984; Smith et al. 1996), but also in how it governs firms (e.g., CIO presence) (Qian et al. 2013). Consequently, future study is needed to assess country-level effects that could impact the explanatory power of our model regarding the interplay between CIOs and boards in terms of AI orientation.

Third, in addition to AI orientation, future study is needed to investigate how the application of AI may impact other facets of firm strategy. Perhaps because AI is a new technology, very few studies have examined the implications of AI applications for firm strategy. As applications of AI mature, future research should direct attention toward the impact of AI on a diverse range of strategic issues central to firm management, such as the effective management of human resources, retaining customers, and managing suppliers. Because the reach of AI could inform how a firm crafts related strategies involving, for example, human resources (Huselid et al. 1997) or marketing (Kim et al. 2016), such interdisciplinary work is important because it could clarify how the interplay between CIOs and boards shapes AI application in terms of the primary and secondary activities necessary to create sustained competitive advantage for firms.

Fourth, our study used a relatively simple conceptualization of CIOs (e.g., whether they were present in the firm or not). While our post hoc operationalization included CTOs, in order to verify the robustness of our results, future research should further examine the effect of top management team composition (Carpenter et al. 2004) and the impact of CIO characteristics such as personality, regulatory focus, etc. (Carter et al. 2011) on AI deployment in organizations. While prior work has differentiated between CIOs who are more or less apt at IT strategy or technology management, it is unclear how this research maps to the introduction of new technologies that have the potential to transform organizations.

Fifth, in our coding of AI orientation, high intelligence robotics became the dominant category (see Figure 2) because the term "high intelligence robotics" is a higher-level AI technology term that includes many AI technologies (e.g., computer vision, speech, NLP, and/or machine learning). However, media coverage and public firms' annual reports are more likely to use application-oriented terms (i.e., intelligent robots) rather than constituent technologies like NLP, machine learning, etc. Given that AI is constantly advancing, future research could examine more advanced ways of measuring AI orientation and discuss its implications for theory and practice.

Sixth, the moderating role of board R&D experience is supported in the main model. However, we found mixed evidence in the robustness testing. One possible explanation for this inconsistency is that board IT-related R&D experience might positively moderate the impact of CIO presence on AI orientation. Due to the limitation of data richness in coding specific R&D experience from the database, we were unable to discriminate among IT-related and other forms of R&D experience. We suggest that future studies further probe the implications of board IT-related R&D experience by assessing whether board IT-related R&D experience might constitute a significant and stable moderating factor.

Conclusion

When and how to effectively develop a strategy to guide AI's deployment has assumed strategic importance for many firms. Drawing on upper echelons theory and the IT management literature, we crafted a novel framework that explains why firms are more likely to develop a strategic AI orientation when they have a CIO and when their boards possess high levels of educational diversity, R&D experience, and AI orientation experience through interlock partners. Further, our post hoc analysis revealed that AI orientation has different drivers than conventional IT orientation and that board-level moderators are more likely to shape AI orientation. Our study provides evidence that AI orientation formulation differs from conventional IT orientation and thus creates a rich empirical foundation for future research seeking to understand the nuances of AI orientation and its strategic implications for firms.

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Appendix A

Summary of Prior IS Research of CIO/Board with Upper Echelons Theory

No.	Source	Category	Focus	IV (Mediator/Moderator)	DV	Key Findings
1	Li, Y., Tan, C. H., Teo, H. H., and Tan, B. C. 2006. "Innovative Usage of Information Technology in Singapore Organizations: Do CIO Characteristics Make a Difference?," *IEEE Transactions on Engineering Management* (53:2), pp. 177-190.	CIO	Personal characteristics; innovative use of IT	Respondent's demographic information (age, tenure, educational level); respondent's personality traits (openness, extraversion, conscientiousness)	Organizational innovative usage of IT (InnoUsg)	• Survey results from 89 CIOs strongly support the proposition. • CIO's personality traits of openness, extraversion, as well as CIO's demographic characteristics of educational level have a strong impact on the innovative organizational use of IT.
2	Cetindamar, D., and Pala, O. 2008. "The Relationship Between CTO and Performance," in *Proceedings of PICMET'08-2008: Portland International Conference on Management of Engineering & Technology*, pp. 42-49.	CTO/TMT	Position; firm performance	The CTO roles; human capital factors (educational background, experience); social capital factors (hierarchical level, TMT membership, extent of networking)	Firm performance (the most global financial performance criteria: profitability)	• Method: survey data from 49 Turkish firms in 2005. • CTO, especially someone who have good educational backgrounds, play a positive role in increasing profitability of firms.
3	Sobol, M. G., and Klein, G. 2009. "Relation of CIO Background, IT Infrastructure, and Economic Performance," *Information & Management* (45:5), pp. 271-278.	CIO	Personal background; firm's financial performance	CIO characteristics (age, gender, education, technical position); IT orientation (utilitarian, strategic); IT infrastructure (technical services, application services, management services, hardware, data services)	Objective financial performance (size, performance, market, knowledge)	• Method: survey data from 123 CIOs. • CIO background and attitude toward IT investment have impact on firms' financial performance. CIO's technical background positively influences the firms' financial performance. The strategic orientation of the CIO can help firms develop a highly profitable profile.
4	Banker, R. D., Hu, N., Pavlou, P. A., and Luftman, J. 2011. "CIO Reporting Structure, Strategic Positioning, and Firm Performance," *MIS Quarterly* (35:2), pp. 487-504.	CIO	Reporting structure; strategic positioning of firms; firm performance	CIO reporting structure; strategic positioning (product differentiation, cost leadership)	Alignment between CIO reporting structure and strategic positioning; firm performance (abnormal stock returns, cash flows from operations)	• Longitudinal data analysis (1990-1993 and 2006). • CIO reporting structure should not be used to gauge the strategic role of IT. • The firm's strategic positioning is the primary determinant of the CIO reporting structure. Besides, the alignment between the strategic positioning of a firm and CIO reporting structure leads to superior firm performance.

No.	Source	Category	Focus	IV (Mediator/Moderator)	DV	Key Findings
5	Khallaf, A., and Skantz, T. 2011. "Does Long Term Performance Improve Following the Appointment of a CIO?," *International Journal of Accounting Information Systems* (12:1), pp. 57-78.	CIO	Position appointment; firm long-term performance	FCIO; LMOVER LEADER; BETA; MV($ millions)	Firm performance (ROA, ROSadp, ROSbdp, MTB, CGS, SGA, OPCOST, SG (sales growth))	• Method: data from 539 announcements in the Lexis–Nexis Wire Service for the period 1987 to 2002 • The CIO appointment has a significant impact on the increase of firm performance, regardless of the "first mover" or "late mover."
6	Blaskovich, J., and Mintchik, N. 2011. "Accounting Executives and IT Outsourcing Recommendations: An Experimental Study of the Effect of CIO Skills and Institutional Isomorphism," *Journal of Information Technology* (26:2), pp. 139-152.	CIO	ICO skills; IT outsourcing recommendations	CIO skills (strong or weak); peer outsourcing actions (no known peer outsourcing or known peer outsourcing)	Participants' recommendations (update current structure; outsource)	• Experiment study . • Accounting executives were more likely to recommend ITO when CIO's skills are weak or questionable.
7	Cetindamar, D., and Pala, O. 2011. "Chief Technology Officer Roles and Performance," *Technology Analysis & Strategic Management* (213:10), pp. 1031-1046.	CTO	Position; firm performance	Fulfilment of CTO roles (Including internal and external); existence of a CTO human capital factor (educational level); social capital factors (hierarchical level, extent of networking)	Company performance (profitability of their company)	• Method: survey data from 49 Turkish firms in 2005. • CTO can play roles in increasing firm's profitability, at the same time, the existence of a distinct managerial technology position can improve profitability.
8	Li, D., Ding, F., and Wu, J. 2012. "Innovative Usage of Information Systems: Does CIO Role Effectiveness Matter?," in *Proceedings of the Pacific-Asia Conference on Information Systems* (p. 81).	CIO	Role characteristics; innovative use of IT	CIO strategic role effectiveness; CIO operational role effectiveness; strategic IS vision	Innovative usage of information systems	• Survey data from 129 CIOs and senior business executives. • CIO role effectiveness has a significant impact on the innovative organizational use of IT. This effect is more salient if the organization has a transform IS vision. • CIO's strategic role effectiveness fully mediates CIO's operational role effectiveness.
9	Li, Y., and Tan, C. H. 2013. "Matching Business strategy and CIO Characteristics: The Impact on Organizational Performance," *Journal of Business Research* (66:2), pp. 248-259.	CIO	Personal characteristics; firms' business strategy	CIO characteristics (age, tenure, educational level, openness, extraversion, consciousness); business strategy (prospector, defender)	Organization's business performance	• Survey data from 81 CIOs/IT managers • The match between business strategy and CIOs' specific repertoires of competencies, experiences, and personality could lead to better organizational performance.
10	Smith, A. L., Bradley, R. V., Bichescu, B. C., and Tremblay, M. C. 2013. "IT Governance Characteristics, Electronic Medical Records Sophistication, and Financial Performance in U.S. Hospitals: An Empirical Investigation," *Decision Sciences* (44:3), pp. 483-516.	CIO	Governance characteristics; adoption of a system	Size; ΔITAp; CIOTRN; CIORptCEO; CIORptDifPos; CIORptDifNeg; SComDifPos; SComDifNeg; CMI; Location; GvrnCtrl; TeachStat; T	EMRSoph; ΔPerformance	• Method: Primary data sources for this study are the HIMSS Analytics Database (2010 more than 5,000 hospitals and 28,000 medical facilities in the U.S., with a focus on health informatics) and the American Hospital Directory (AHD) (2010) provides online annual clinical, operational, and financial performance data on more than 6,000 hospitals (both active and inactive) CIO reports to CEO, CIO turnover is low, and the IT steering committee is present are more likely to have a sophisticated electronic medical record system, thus leading to improved financial performance. • Results underscore the importance of continuity in the CIO position.

No.	Source	Category	Focus	IV (Mediator/Moderator)	DV	Key Findings
11	Lin, T. C., Ku, Y. C., and Huang, Y. S. 2014. "Exploring Top Managers' Innovative IT (IIT) Championing Behavior: Integrating the Personal and Technical XContexts," *Information & Management* (51:1), pp. 1-12.	CIO/ TMT	Personal characteristics; innovative IT championing behavior	Openness to experience; optimum stimulation level; information literacy; involvement; IIT absorptive capacity	IT championing behavior	• Survey data from 130 top managers. • The top managers' absorptive capacity, involvement, and optimum stimulation level could enhance their innovative information technology championing behavior.
12	Ding, F., Li, D., and George, J. F. 2014. "Investigating the Effects of IS Strategic Leadership on Organizational Benefits from the Perspective of CIO Strategic Roles," *Information & Management* (51:7), pp. 865-879.	CIO	IS strategic leadership; organizational performance	IS strategic leadership (IS strategist role leadership; nusiness strategist role leadership); IS quality (systems quality, information quality; service quality); IS vision	Organizational benefits (strategic benefits, managerial/ controlling effectiveness)	• Method: Survey data from 110 matched pairs of CIOs and business executives within organizations. • CIO strategic leadership significantly affects both organizational benefits and information system quality. The IS quality mediates this effect. Besides, IS vision is found to moderate the relationship between CIO strategic leadership and IS quality.
13	Hu, Q., Yayla, A. A., and Lei, Y. 2014. "Does Inclusion of CIO in Top Management Team Impact Firm Performance? Evidence from a Long-Term Event Analysis," in *Proceedings of the 47th Hawaii International Conference on System Sciences*, pp. 4346-4355.	CIO	Position; firm performance	Size ($ millions: market value, total assets) per-event performance (Tobin's Q; ROA); environment dynamism; time period	Yearly abnormal change; cumulative abnormal change	• Long-term event study methodology. • Firm-level secondary data analysis (ExecuComp database). • CIO in TMT has a significant positive effect on firm performance. This positive effect is larger for firms in dynamic environments.
14	Taylor, J., Sahym, A., and Vithayathil, J. 2015. "Do Powerful Technology Leaders Make a Difference in Firm Performance?," in *Proceedings of the 48th Hawaii International Conference on System Sciences*, pp. 4502-4512.	CIO	Position; firm performance	Technology leader inclusion in the management upper echelon	Selling, general and administrative expenses post 3 years, post 5 years;return on assets, post 3 years, post 5 years; sales post 3 years, post 5 years; Tobin's q post 3 years, post 5 years	• Method: Longitudinal data analysis (1992-2013) In total just under 250,000 company-year combinations are included in this study. • Firms with a powerful senior technical leader perform better on forward-oriented dimensions of value, such as sales growth and Tobin's Q.
15	Bergeron, F., Croteau, A. M., Uwizeyemungu, S., and Raymond, L. 2015. "IT Governance Theories and the Reality of SMEs: Bridging the Gap," in *Proceedings of the 48th Hawaii International Conference on System Sciences*, pp. 4544-4553.	ITG	IT governance; business value	SME owner-manager's characteristics (IT competencies, interpersonal and managerial competencies); SEM's external links (outside directors, institutional pressures, external IT partnerships); key employees' characteristics (IT technical skills, business competencies, values/ beliefs/norms); IT governance mechanisms (structures, processes, participants)	IT business value	• Summary paper. • Create a conceptual framework for ITG in small and medium-sized enterprises (SMEs).

No.	Source	Category	Focus	IV (Mediator/Moderator)	DV	Key Findings
16	Xu, F., Zhan, H., Huang, W., Luo, X. R., and Xu, D. 2016. "The Value of Chief Data Officer Presence on Firm Performance," in *Proceedings of the Pacific Asia Conference on Information Systems*, p. 213.	Chief data officer (CDO).	Position; firm performance	Average (return on assets); average (sales growth); average (market to book ratio); POST(the difference in performance before and after the appointment)	CDO presence; firm performance (accounting-based performance and market-based performance)	• Three data sources for 68 firms with a CDO position: – 29 US firm data from announcements in the Lexis–Nexis Wire Service for the period – collect information of 41 CDOs who are not formally be titled as CDOs but functional as a CDO through their public statement – utilize four-digit Standard Industry Classification (SIC) scheme as a benchmark to select the control group for each incident firm • The return on assets (ROA) is positively associated with CDO appointment, however, market to book ratio (M/B) is negatively associated with CDO appointment. • Firms with CDO have better financial performance than their peers without CDO.
17	Caluwe, L., and De Haes, S. 2019. "Board Level IT Governance: A Scoping Review to Set the Research Agenda," *Information Systems Management* (36:3), pp. 262-283.	ITG	IT governance	NA	NA	• Scoping review paper. • Involvement of boards in governing digital assets is low.

Appendix B

Summary of CIO-Related Research in Top IS Journals

No.	Source	Category	Focus	IV (Mediator/Moderator)	DV	Theory	Key Findings
1	Enns, H. G., Huff, S. L., and Higgins, C. A. 2000. "CIO Lateral Influence Behaviors: Gaining Peers' Commitment to Strategic Information Systems," in *Proceedings of the 21st International Conference on Information Systems*, pp. 457-460.	CIO	Personal characteristics; peer commitment	Rational persuasion; conclusion; personal appeal; ingratiation; exchange; coalition; pressure	Influence outcome (commitment, compliance, or resistance)	Null	• Dyadic survey data was used. 69 pairs of CIO-CEO. • The influence behavior rational persuasion and personal appeal have a significant impact on peer commitment. Moreover, exchange and pressure are significantly related to peer resistance.
2	Karimi, J., Bhattacherjee, A., Gupta, Y. P., and Somers, T. M. 2000. "The Effects of MIS Steering Committees on Information Technology Management Sophistication," *Journal of Management Information Systems* (17:2), pp. 207-230.	ITM	Information technology management, information technology management sophistication, steering committees	Steering group; policy committee; IT board	IT planning; IT control; IT organization; IT integration	Organizational learning theory	• Survey data from 213 firms in the financial services sector (e.g., banks, insurance companies). • The presence and roles of IT steering committees are significantly associated with the level and nature of IT management sophistication within firms.
3	Tai, L. A., and Phelps, R. 2000. "CEO and CIO Perceptions of Information Systems Strategy: Evidence from Hong Kong," *European Journal of Information Systems* (9:3), pp. 163-172.	CIOs	Positions	CEO&CIO IT vision; CEO&CIO technology perceptions; CEO/CIO KM perceptions	Perceived IT success	Null	• Survey data from 139 reposes (63 CEOs and 76 CIOs/IT executives). • There are similarities of views in three main dimensions of perception: IT vision, organizational IS issues and IT support for knowledge management. • National culture, industry type and the chief executive/IT executive relationship causes differences in IT perception.
4	Chatterjee, D., Richardson, V. J., and Zmud, R. W. 2001. "Examining the Shareholder Wealth Effects of Announcements of Newly Created CIO Positions," *MIS Quarterly* (25:1), pp. 43-70.	CIO	Position; firm performance	External; IT-driven transformation level; time	CARs	Null	• Event study method. • For firms competing in industries undergoing IT-driven transformation, an announcement of newly created CIO positions does indeed provoke positive reactions from the marketplace.
5	Bassellier, G., Reich, B. H., and Benbasat, I. 2001. " Information Technology Competence of Business Managers: A Definition and Research Model," *Journal of Management Information Systems* (17:4), pp. 159-182.	ITM	Explicit and tacit knowledge, information technology competence, information technology management	Explicit IT knowledge (technology, application, systems development, management of IT, access to IT knowledge); tacit IT knowledge (experience, cognition)	IT competence; attitudes toward line technology leaderships; subjective norms; perceived behavioral control; intentions toward line technology leadership; line technology leadership	Theory of planned behavior	• Literature review works. • There are two chiefly two behaviors that indicated the outcomes expected from IT-competent business managers, including (1) an increased willingness to form partnerships with IT people and (2) an increased propensity to lead and participate in IT projects

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6	Enns, H. G., Huff, S. L., and Golden, B. R. 2001. "How CIOs Obtain Peer Commitment to Strategic IS Proposals: Barriers and Facilitators," *The Journal of Strategic Information Systems* (10:1), pp. 3-14.	CIOs	Behavior; information systems; strategic planning level	Peer technical background	Influence behaviors; Influence outcomes;	Null	• Method: focus study and survey. • Peer commitment barriers and facilitators consist of the firm's external and internal IS/T environment, appropriateness of the IS/T initiatives, peer relationships, the ability to use the peer's preferred influence behaviors, and post-commitment implementation realities.
7	Kearns, G. S., and Lederer, A. L. 2003. "A Resource- Based View of Strategic IT Alignment: How Knowledge Sharing Creates Competitive Advantage," *Decision Sciences* (34:1), pp. 1-29.	IS planning	Competitive advantage, information systems planning, knowledge sharing, resource-based view, strategic planning, and structural equation modeling	Information intensity of the value chain; the CIO participates in business planning; the CEO participates in IT planning; the IT plan reflects the business plan; the business plan reflects the IT plan	IT is used for competitive advantage	Resource-based theory	• Survey data from 161 CIOs responses. • Information intensity is an important keys to strategic IT alignment.
8	Schwarz, A., and Hirschheim, R. 2003. "An Extended Platform Logic Perspective of IT Governance: Managing Perceptions and Activities of IT," *The Journal of Strategic Information Systems* (12:2), pp. 129-166.	IT governance	Information technology management; oil and gas industry; governance platform logic model; relation-based management	IT capabilities; relational architectures; integration architecture; success metrics; IT–business unit relationship	Null	Null	• Case study. • When organizations pay more attention to implementing a sound IT governance strategy, which helps senior executives to manage the IT-related activities and the perceptions between IT and the rest of the organization, which promotes IT organization more successful.
9	Pawlowski, S. D., and Robey, D. 2004. "Bridging User Organizations: Knowledge Brokering and the Work of Information Technology Professionals," *MIS Quarterly* (28:4), pp. 645-672.	IT professionals	Boundary spanning, organizational communication, organizational learning, IS skill requirements, IT professionals, knowledge broker, internal knowledge transfer	Position of IT professionals; shared systems as boundary objects; brokering practices	Consequences of knowledge brokering	Boundary spanning; situated learning theory	• Qualitative analysis of interviews conducted with 23 IT professionals and business users in a large manufacturing and distribution company. • Structural conditions, including decentralization and a federated IT management organization, and by technical conditions, can affect brokering practices. Brokering practices consist of gaining permission to cross organizational boundaries, surfacing and challenging assumptions made by IT users, translation and interpretation, and relinquishing ownership of knowledge. Consequences of brokering refers to the transfer of both business and IT knowledge across units in the organization.

No.	Source	Category	Focus	IV (Mediator/Moderator)	DV	Theory	Key Findings
10	Johnson, A. M., and Lederer, A. L. 2005. "The Effect of Communication Frequency and Channel Richness on the Convergence Between Chief Executive and Chief Information Officers," *Journal of Management Information Systems* (22:2), pp. 227-252.	CIO	CEO/CIO communication, channel richness, communication frequency, managerial communication, media richness, strategic grid.	Communication frequency; communication channel richness	CEO/CIO convergence on the current role of IT; CEO/CIO convergence on the future role of IT: managerial support; CEO/ CIO convergence on the future role of IT: differentiation; CEO/CIO convergence on the future role of IT: enhancement; IS financial contribution	Communication theory; media richness theory	• Survey data from survey of 202 pairs of CEOs and CIOs. • Convergence (i.e., mutual understanding) between an organization's CEO and CIO plays roles in successfully exploiting information technology. More frequent communication can predict convergence about the current role, differentiation future role, and enhancement of future role. The use of richer channels can predict convergence about the differentiation of future roles.
11	Kearns, G. S., and Sabherwal, R. 2006. "Strategic Alignment Between Business and Information Technology: A Knowledge-Based View of Behaviors, Outcome, and Consequences," *Journal of Management Information Systems* (23:3), pp. 129-162.	ITB	Business effect of information technology, business–IT strategic alignment, information technology planning, IT project planning, knowledge management.	Organizational emphasis on knowledge management; centralization of IT decisions	Top manager' knowledge of IT; business-IT strategic alignment; business impact of IT; firm performance	Knowledge-based theory	• Data from a survey of 274 senior information officers . • Organizational emphasis on knowledge management and centralization of IT decisions can affect top managers' knowledge of IT. Quality of IT project planning and implementation problems in IT projects mediate the relationship between business-IT strategic alignment and business effect of IT.
12	Watts, S., and Henderson, J. C. 2006. "Innovative IT Climates: CIO Perspectives," *The Journal of Strategic Information Systems* (15:2), pp. 125-151.	CIOs	Innovative IT; technology leadership; climate; CIO	Innovative CIO practice	Innovative IT climate	Organizational climate theory	• Data were collected by means of face-to-face interviews with 36 CIO's of companies identified as highly innovative. • CIO can play roles in is driving business innovation. The dimensions of innovative IT climates can be characterized by using a theoretical model based on the climate literature.
13	Liang, H., Saraf, N., Hu, Q., and Xue, Y. 2007. "Assimilation of Enterprise Systems: The Effect of Institutional Pressures and the Mediating Role of Top Management," *MIS Quarterly* (31:1), pp. 59-87.	Enterprise management systems TMB/TMP; top management	Top management; assimilation of enterprise system	Top management beliefs; top management participation; mimetic pressures; coercive pressure; normative pressures	Assimilation	Institutional theory and the influence of top management	• Survey data. • Mimetic pressures positively affect top management beliefs, which then positively affects top management participation in the ERP assimilation process. • Coercive pressures positively affect top management participation without the mediation of top management beliefs. • Normative pressures directly influence ERP usage.
14	Zhang, J., and Faerman, S. R. 2007. "Distributed Leadership in the Development of a Knowledge Sharing System," *European Journal of Information Systems* (16:4), pp. 479-493.	IS leadership	IS leadership; project leadership; distributed leadership; knowledge sharing	Leadership roles: provide a vision; reconstruct organizational structures and processes (including technology adaptation); coordinate knowledge sharing	NA	Null	• Case study: Semi-structured interviews were held with 19 participants across different units, offices, divisions and organization. • Different styles leaders play different roles: project leader who is knowledgeable and persistent can play role in the spearheading and coordinating; perceptive and collaboration-inclined executives can play roles in supporting and steering; knowledge champions can take roles in knowledge sharing and momentum driving.

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15	Preston, D. S., Leidner, D. E., and Chen, D. 2008. "CIO Leadership Profiles: Implications of Matching CIO Authority and Leadership Capability on IT Impact," *MIS Quarterly Executive* (7:2).	CIO	Role characteristics; leadership profile; IT investment payoff	CIO leadership profile (IT orchestrator, IT advisor, IT laggard, IT mechanic); CIO attributes (strategic knowledge, interpersonal skills); CIO Integration with top management (percentage reporting to the CEO, percentage a member of top management team); IT commitment (IT resource, strategic IT vision)	IT contribution level	Null	• Method: six semi-structured interviews with industry CIOs and matched-pairs survey from 174 US organizations from different industries. • The CIO leadership profile, which consists of strategic decision-making authority, strategic leadership capability, has a significant impact on the IT investment payoff. • CIO leadership profile can be represented as an IT orchestrator, IT laggard, IT advisor, and IT mechanic. Over time, there will be a shift to IT orchestrators, and CIO lacking the necessary characteristics should plan to acquire them.
16	Sidorova, A., Evangelopoulos, N., Valacich, J. S., and Ramakrishnan, T. 2008. "Uncovering the Intellectual Core of the Information Systems Discipline," *MIS Quarterly* (32:3), pp. 467-482.	IS identity	Identity, IS research issues, IS research agenda, organizational identity, latent semantic analysis	NA	NA	Null	• Review paper. • Information systems academic discipline has maintained a relatively stable research identity that focuses on how IT systems are developed and how individuals, groups, organizations, and markets interact with IT.
17	Preston, D. S., Chen, D., and Leidner, D. E. 2008. "Examining the Antecedents and Consequences of CIO Strategic Decision Making Authority: An Empirical Study," *Decision Sciences* (39:4), pp. 605-642.	CIOs	Position; firm performance	Organizational climate; organizational support for IT; CIO structural power; CIO/TMT partnership; CIO strategic effectiveness	CIO strategic decision-making authority; IT contribution to firm performance	Theory of managerial discretion	• Data from a cross-industry sample of 174 matched pairs of CIOs and top business executives. • There are four factors that can directly affect the CIO's level of strategic decision making authority within the organization, including organizational climate, organizational support for IT, the CIO's structural power, the CIO's level of strategic effectiveness, and a strong partnership between the CIO and top management team.
18	Mishra, A. N., and Agarwal, R. 2010. "Technological Frames, Organizational Capabilities, and IT Use: An Empirical Investigation of Electronic Procurement," *Information Systems Research* (21:2), pp. 249-270.	IT use	Electronic procurement; B2B electronic markets; technological frames; benefits frame; threat frame; adjustment frame; organizational capabilities; technological opportunism; technological sophistication; sensemaking	Technological frames (benefits frame, threat frame, adjustment frame); organizational capabilities (technological opportunism, technological sophistication)	The extent of B2B markets use	IT use in organizations; sensemaking, technological frames, and organizational action; organizational capabilities and innovation	• Survey data collected from 292 firms. • There are interactions between three technological frames—benefits frame, threat frame, and adjustment frame, and two organizational capabilities—technological opportunism and technological sophistication, and their relationship with the use of B2B electronic markets in firms.
19	Chen, D. Q., Preston, D. S., and Xia, W. 2010. "Antecedents and Effects of CIO Supply-Side and Demand-Side Leadership: A Staged Maturity Model. *Journal of Management Information Systems* (27:1), pp. 231-272.	CIOs	Chief information officer, exploitation, exploration, IT functional impact, IT leadership, staged maturity model, strategic value of IT, structural equation modeling, survey research	CIO human capital; CIO structural power; organizational support for IT	Supply-side leadership, demand-side leadership, contribution to firm efficiency, contribution to strategic growth)	Strategic leadership theory, capital theory	• Survey data from 285 business executive respondents across 174 organizations. • The chief information officer (CIO) play the traditional supply-side leadership role that focuses on exploiting existing IT competencies to support known business needs, it also can play the demand-side leadership role that focuses on exploring new IT-enabled business opportunities that result in competitive advantage.

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20	Leidner, D. E., Preston, D., and Chen, D. 2010. "An Examination of the Antecedents and Consequences of Organizational IT Innovation in Hospitals," *The Journal of Strategic Information Systems* (19:3), pp. 154-170.	IT innovation	IT leadership; IT innovation; strategic management of IT; competitive impacts of IT; chief information officer; top management team; structural equation modeling; questionnaire surveys	CIO strategic leadership; TMT attitude toward IT; hospital climate	Hospital IT innovation; IT impact; hospital performance	IT innovation theory	• Data from interviews with 10 CIOs of hospitals and healthcare systems. • The CIO's strategic leadership and the TMT's attitude toward IT play a key role in influencing IT innovation, but the influence of a hospital's climate on organizational IT innovation is contingent upon the CIO's level of strategic leadership.
21	Huang, R., Zmud, R. W., and Price, R. L. 2010. " Influencing the Effectiveness of IT Governance Practices through Steering Committees and Communication Policies," *European Journal of Information Systems* (19:3), pp. 288-302.	IT governance	IT governance; IT use; steering committee; senior management involvement; communication practices; small- and medium-sized enterprise (SME)	Governance structure; formal senior management involvement; informal senior management involvement; IT governance communication channels; primary IT governance communication channels	Efficiency of IT use; breadth of current IT use; breadth of potential IT use	Null	• Qualitative data by interviewing with CIO or divisional VPs. • While each firm evidenced a centralized IT governance posture, differences in these firms' IT steering committees and governance-related communication policies can explain differences in the firms' IT use outcomes.
22	Lu, Y., and Ramamurthy, K. 2010. "Proactive or Reactive IT Leaders? A Test of Two Competing Hypotheses of IT Innovation and Environment Alignment," *European Journal of Information Systems* (19:5), pp. 601-618.	IT leaders	IT innovation; industry dynamism; IT innovation-environment alignment	Group; performance outcome; time; past performance; firm size; rank index; IT spending; industry dynamism; industry munificence; industry concentration; industry sector	Industry average adjusted performance score	Real options theory	• Method: quasi-experiment design by using archival data of IT leaders over a time frame of 6 years; a longitudinal analysis of the performance change trajectories of proactive and reactive IT leaders over time. • Proactive IT leaders that lead in innovation with IT in relatively stable environments prefer to consistently outperform reactive IT leaders in overall performance, allocative efficiency, and cost efficiency in the management process. But the reactive IT leaders that prefer the intensive deployment of IT innovation in dynamic environments are likely to gain a cost advantage in the production and operation process over time.
23	Carter, M., Grover, V., and Thatcher, J. B. 2011. "The Emerging CIO Role of Business Technology Strategist," *MIS Quarterly Executive* (10:1).	CIO	Role characteristics; CIO competence of IT management	CIO roles; formal power; IT leader's credibility	Business technology strategist	Null	• Survey method. • CIO role of business technology strategist is most strongly related to the CIO's formal power and to his or her skill in absorbing and disseminating relevant information (the information role). A technical background is more important for the performance of the decisional and interpersonal IT management.
24	Kettinger, W. J., Zhang, C., and Marchand, D. A. 2011. "CIO and Business Executive Leadership Approaches to Establishing Company-Wide Information Orientation," *MIS Quarterly Executive* (10:4).	CIO	Information orientation	Information technology practices; information management practices; information behaviors and values	Information orientation	Null	• Qualitative study. • CIOs ought to create business value rather than manage IT utility, which play a role in helping companies to create a strong information usage culture. • The guidelines were proposed to help CIOs to position their leadership challenges in introducing or sustaining the companies' information orientation in initiatives and recommends specific leadership approaches depending on CIOs' particular situations.

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25	Peppard, J., Edwards, C., and Lambert, R. 2011. "Clarifying the Ambiguous Role of the CIO," *MIS Quarterly Executive* (10:1), pp. 31-44.	CIO	Role characteristics; CIO's value to firms	Degree of charity of CIO role; understanding the need for a CIO; performance of the CIO; a CIO's ability to deliver against senior management expectations; the evolution of the CIO role	Null	Null	• The interview method was used. • The ambiguity of the CIO's value is rooted in the fact that there are five distinct CIO roles: utility IT director, evangelist CIO, innovator CIO, facilitator CIO, and agility IT CIO. • The criticality of information and technology determines the CIO's value for a particular organization for competitive differentiation and the maturity of information leadership capabilities.
26	Mithas, S., Ramasubbu, N., and Sambamurthy, V. 2011. "How Information Management Capability Influences Firm Performance," *MIS Quarterly* (35:1), pp. 237-256.	Information management capability	Information management capability, information technology, IT capability, customer management capability, process management capability, performance management capability, organizational capital, firm performance, performance excellence, business excellence, resource-based view	Information management capability (performance capability, customer capability, process management capability)	Customer-focused results; financial results; human resource results; organizational effectiveness results	Information management capability	• Data from the examiners' evaluations of firms and intra-organizational units in the business group for the period 1999-2000. • A conceptual model was proposed in this study, which matches IT-enabled information management capability with three important organizational capabilities, including customer management capability, process management capability, and performance management capability. • Three capabilities can play roled in mediating the relationship between information management capability and firm performance. • Information management capability plays a significant function in enhancing other firm capabilities for customer management, process management, and performance management. In turn, these capabilities favorably affect customer, financial, human resources, and organizational effectiveness measures of firm performance.
27	Bell, G. G., Lai, F., and Li, D. 2012. "Firm Orientation, Community of Practice, and Internet-Enabled Interfirm Communication: Evidence from Chinese Firms," *The Journal of Strategic Information Systems* (21:3), pp. 201-215.	Community of practice	Firm orientation; learning orientation; customer orientation; competitor orientation; community of practice; Internet-enabled interfirm communication	Learning orientation; customer orientation; competitor orientation; domestic tie; foreign tie; size; age	Community of practice; Internet-enabled interfirm communication	Null	• Data collected from 307 international trade firms in the Beijing area. • The internal community of practices and customer orientation will play a role in directly driving Internet-enabled interfirm communication, but it plays a role in indirectly driving competitor orientation and learning orientation. Learning orientation and competitor orientation can influence the internal community of practice, but customer orientation can not affect it.

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28	Karahanna, E., and Preston, D. S. 2013. "The Effect of Social Capital of the Relationship Between the CIO and Top Management Team on Firm Performance," *Journal of Management Information Systems* (30:1), pp. 15-56.	ICOs, TMT	Chief information officer, cognitive social capital, financial performance, information systems, IS leadership, relational social capital, social capital, strategic alignment, structural social capital, top management team	Structural social capital; cognitive social capital; relational social capital; IS strategic alignment	Firm performance	Social capital theory, social identity theory, information processing theory	• Hospital data (other than financial performance) were collected via a matched-pair survey of CIOs and a TMT peer executive. Financial performance data were derived from the American Hospital Directory. • IS alignment plays a significant role in influences the firm's financial performance, which mediates the relationship between CIO-TMT social capital and performance. Besides, cognitive and relational social capital will affect information systems strategic alignment, yet structural social capital exerts will influence through its effects on cognitive social capital.
29	Lim, J. H., Stratopoulos, T. C., and Wirjanto, T. S. 2013. "Sustainability of a Firm's Reputation for Information Technology Capability: The Role of Senior IT Executives," *Journal of Management Information Systems* (30:1), pp. 57-96.	IT executives	External legitimacy, institutional theory, internal legitimacy, IT capability reputation, IT executives, IT strategic leadership, reciprocity, structural power	IT capability reputation; structural power; IT-related expert power	Appreciation, reward, and continuity of IT strategic leadership	Institutional theory	• Panel data for 1,326 large U.S. firms from a wide spectrum of industries over a 13-year period (1997–2009). • IT executives with greater structural power (e.g., higher job titles) or IT-related expert power (e.g., IT-related education or experience) are more likely to attract public recognition for their firm's IT capability. • The continuity in IT strategic leadership is positively associated with the firm's ability to sustain its IT capability reputation.
30	Smith, A. L., Bradley, R. V., Bichescu, B. C., and Tremblay, M. C. 2013. "IT Governance Characteristics, Electronic Medical Records Sophistication, and Financial Performance in U.S. Hospitals: An Empirical Investigation," *Decision Sciences* (44:3), pp. 483-516.	CIO	Governance characteristics; adoption of a system	Size; ΔITAp; CIOTRN; CIORptCEO; CIORptDifPos; CIORptDifNeg; SComDifPos; SComDifNeg; CMI; Location; GvrnCtrl; TeachStat; T	EMRSoph; ΔPerformance	Upper echelons theory	• Method: Primary data sources for this study are the HIMSS Analytics Database(2010; more than 5,000 hospitals and 28,000 medical facilities in the U.S., with a focus on health informatics) and the American Hospital Directory (AHD) (2010; provides online annual clinical, operational, and financial performance data on more than 6,000 hospitals (both active and inactive) CIO reports to CEO, CIO turnover is low, and the IT steering committee is present are more likely to have a sophisticated electronic medical record system, and thus leading to improved financial performance. • Results underscore the importance of continuity in the CIO position.
31	Saraf, N., Liang, H., Xue, Y., and Hu, Q. 2013. "How Does Organizational Absorptive Capacity Matter in the Assimilation of Enterprise Information Systems?," *Information Systems Journal* (23:3), pp. 245-267.	Enterprise information systems	IT assimilation, enterprise systems, ERP assimilation, institutional influences, absorptive capacity, organizational learning	Mimetic; TMB; Coercive; TMP; Assimilation; Normative	PACAP; RACAP	Absorptive capacity	• 77 survey responses from managers of Chinese companies. • While both PACAP and RACAP have a positive direct influence on assimilation, PACAP positively moderates the impact of mimetic (institutional) pressures, but not normative (institutional) pressures, on assimilation; whereas RACAP positively moderates the impact of normative pressures, but not mimetic pressures, on assimilation.

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32	Yayla, A. A., and Hu, Q. 2014. "The Effect of Board of Directors' IT Awareness on CIO Compensation and Firm Performance," *Decision Sciences* (45:3), pp. 401-436.	CIOS	Position; firm performance	IT awareness of the broad; IT intensity of industry	CIO compensation structure; firm performance	Agency theory; resource dependence theory, strategic alignment theory	• 433 compensation data reports over the 1994–2005 time periods: compensation data from Standard & Poor's Execu Comp database; industry IT intensity data, we used the capital-spending database compiled by the U.S. BEA. • Boards' IT awareness plays the important role in shaping CIO compensation and improving firm performance. Specifically, boards with functional area knowledge—or higher IT awareness in this case—can more effectively monitor and better incentivize executives, and consequently lead to better firm performance.
33	Wu, S. P. J., Straub, D. W., and Liang, T. P. 2015. "How Information Technology Governance Mechanisms and Strategic Alignment Influence Organizational Performance: Insights from a Matched Survey of Business and IT Managers," *MIS Quarterly* (39:2), pp. 497-518.	IT managers	Strategic alignment; IT governance; organizational performance	IT governance mechanisms; IS strategic alignment	Organizational performance	Resource-based view	• A matched survey from 131 Taiwanese companies. • IT governance mechanisms have a positive and significant impact on strategic alignment. Further, strategic alignment positively influences the organizational performance.
34	Sharma, S., and Rai, A. 2015. "Adopting IS Process Innovations in Organizations: The Role of IS Leaders' Individual Factors and Technology Perceptions in Decision Making," *European Journal of Information Systems* (24:1), pp. 23-37.	IS leaders	Organizational decision making; diffusion of innovation; information technol ogy adoption; IS innovation; IS leaders; CASE	Individual factors: hierarchical position of the IS leader; job tenure of the IS leader Technological factors: perceived relative advantage; perceived complexity	The CASE adoption decision in an organization	Theory of reasoned action; organizational decision-making theory	• Survey data from 360 the head of ISD. • IS leaders' hierarchical position and their job tenure significantly differentiate CASE adopters from non-adopters. Besides, relative advantage has two dimensions, such as perceived efficacy advantage and perceived efficiency advantage.
35	Masli, A., Richardson, V. J., Watson, M. W., and Zmud, R. W. 2016. "Senior Executives' IT Management Responsibilities: Serious IT-Related Deficiencies and CEO/CFO Turnover. *MIS Quarterly* (40:2), pp. 687-708.	IT management	IT management, executive responsibilities, managerial delegation, executive turnover	IT weakness variables(IT control oversight–internal; IT capability; software development; IT architecture; IT control oversight–external)	CEO and CFO turnover	Senior executive roles in modern organizations, managerial delegation, and the salience of IT within the post-SOX financial reporting context	• Collected the data used to assess research hypotheses from Audit Analytics (e.g., SOX 404 reports), annual Compustat (financial statement variables), SEC financial reports (e.g., DEF 14-A; 10-K), Execucomp (CEO and CFO variables), and a variety of sources including the Million Dollar Database, Lexis-Nexis, ZoomInfo, and Google Search. • CEOs should play roles in IT management responsibilities; CFOs should play roles in for demand-side IT management responsibilities.

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36	Dawson, G. S., Denford, J. S., Williams, C. K., Preston, D., and Desouza, K. C. 2016. "An Examination of Effective IT Governance in the Public Sector Using the Legal View of Agency Theory," *Journal of Management Information Systems* (33:4), pp. 1180-1208.	IT governance	Agency theory, CIO, IT governance, IT in public sector, public sector, steering committee	Legislative oversight committee; business centric IT steering committee; independent CIO; business unit led funding	IT department performance; overall state performance	Agency theory	• Data were collected from three archival sources: the National Association of State CIOs (NASCIO) (LOC, independent CIO, business-centric IT steering committee, and fee-for-service funding), the Grading the States project from the Center for Digital Government (CDG) (IT departmental performance), and the Government Performance Project (GPP) from the Pew Center on the States (PCS) (state performance). • The study suggests that by shifting from the control-oriented view of governance in the private sector to a more mediating view in the public sector, important practices may be portable between the public and private sector, despite their widely differing structures.
37	Benlian, A., and Haffke, I. 2016. "Does Mutuality Matter? Examining the Bilateral Nature and Effects of CEO–CIO Mutual Understanding," *The Journal of Strategic Information Systems* (25:2), pp. 104-126.	CIOs	Position	CEO's understanding of CIO; CIO's understanding of CEO	Quality of collaboration IT contribution; perceptual congruence facets actual agreement; perceived agreement	Null	• Survey data from 1000 CEO–CIO pairs from randomly selected companies in Germany. • Both executives' actual opinions on important business and IT topics are more similar than both perceive them to be. Accordingly, perceptions of each other's opinions are negatively biased away from their real opinions. CIOs' understanding of their CEO plays a more important role in predicting the quality of CEO–CIO collaboration than CEOs' understanding of their CIO.
38	Shao, Z., Feng, Y., and Hu, Q. 2016. "Effectiveness of Top Management Support in Enterprise Systems Success: A Contingency Perspective of Fit Between Leadership Style and System Life-Cycle," *European Journal of Information Systems* (25:2), pp. 131-153.		Enterprise systems lifecycle; contingency theory; transactional leadership; transformational leadership; top management; ERP success; multi-case study	Leadership characteristic; leadership style	Success indicators	Strategic leadership; enterprise systems life-cycle	• Case study: interview with 27 IS leaders . • Different leadership style fits different stages of IS adoptions. Transformational leadership can play roles in the adoption phase, transactional leadership can play roles in the implementation phase. But two variations of combined transformational and transactional leadership styles can play the most effective role in the assimilation and extension phases.
39	Tumbas, S., Berente, N., and vom Brocke, J. 2017. "Three Types of Chief Digital Officers and the Reasons Organizations Adopt the Role," *MIS Quarterly Executive* (16:2).	Chief Digital Officers	CDOs positions	Digital innovation; data analytics; customer engagement	Types of CDOs	Null	• SE paper. • Exploratory interviews with 35 CDOs from industries spanning automotive, financial services, healthcare, software, publishing, governmental and not-for-profit organizations in a variety of countries.
40	Marabelli, M., and Galliers, R. D. 2017. "A Reflection on Information Systems Strategizing: the Role of Power and Everyday Practices," *Information Systems Journal* (27:3), pp. 347-366.	Information systems strategy	Information systems strategizing, strategy-as-practice, exploration, exploitation, power, case vignettes	NA	NA	NA	• Review paper. • Strengths of IS strategizing include an account given to the relevance of tensions between planned and executed strategy, and related tradeoffs like rigidity and flexibility, formal and informal strategizing and the exploitation of static resources vis à vis the exploration of novel capabilities.

No.	Source	Category	Focus	IV (Mediator/Moderator)	DV	Theory	Key Findings
41	Chan, R. Y., and Ma, K. H. 2017. "Impact of Executive Compensation on the Execution of IT-Based Environmental Strategies under Competition," *European Journal of Information Systems* (26:5), pp. 489-508.	CEO	CEO compensation; agency theory; IT-based environmental strategies; competitive intensity	Concerned the percentages of CEO fixed pay (FIXED); bonuses (BONUS); stock options (OPTION); competitive intensity (COMP)	Green IT strategies (GITS); IT-enabled green strategies (ITEGS	Agency theory, contingent natural resource-based theory	• Two sources of data: (1) 263 survey data for firms' IT-based environmental strategies and the level of competitive intensity they faced; (2) archival data. • First, CEO fixed pay and bonus negatively affect the execution of the two types of IT-based environmental strategies, green IT strategies and IT-enabled green strategies, whereas CEO stock option positively influences such execution. The moderation analysis further highlights that while competitive intensity reinforces the positive impact of CEO stock options on both strategies, it weakens the negative impact of CEO fixed pay and bonus only on IT-enabled green strategies. Second, a highly competitive operating setting represents an ideal setting for using stock option to motivate CEOs to execute these two strategies. At the minimum, this setting helps mitigate the deterrent effect of fixed pay and bonus on CEOs' execution of IT-enabled green strategies. Third, firms should align their CEO compensation package with the characteristics of high uncertainty and long payback periods of IT-based environmental strategic endeavors and the competitive conditions they face.

No.	Source	Category	Focus	IV (Mediator/Moderator)	DV	Theory	Key Findings
42	Xue, L., Yang, K., and Yao, Y. 2018. "Examining the Effects of Interfirm Managerial Social Ties on IT Components Diversity: An Agency Perspective," *MIS Quarterly* (42:2), pp. 679-694.	IMSTs	Interfirm managerial social ties, agency theory, IT components diversity, IT supplier, social embeddedness	Sup_Num; Tie_Sup; Tie_Div; Divi_Num; Diver; Size; Adj_ROA; R&D; Age; IT_Inten; Avg_Sup	IT components diversity (ITCompo_ Div)	Agency theory	• Data from three sources: (1) data on IT suppliers and IT components diversity is collected from the Computer Intelligence (CI) Technology database, (2) the data on IMSTs are collected from the BoardEx database by Management Diagnostic Limited,(3) the data on firms' and suppliers' characteristics are collected from the Compustat database. • This study indicated that interfirm managerial social ties (IMSTs) have influences on IT components diversity and the conditions. The firms will meet higher IT components diversity internally if firms are connected with more IT suppliers through IMSTs and IMSTs are more evenly distributed over connected IT suppliers. Besides, the positive relationship between IMSTs and IT components diversity is stronger in IT infrastructure components than in IT applications/services components. It is stronger in the firms with centralized IT decision making than in the firms with decentralized IT decision making. • There is a potential dark side of IMSTs from the perspective of agency theory. If corporate decision makers with broader and more diversified social connections with IT suppliers, they are more likely to recruit IT suppliers in a way that escalates the diversity of IT components within the firm and jeopardizes the overall IT standardization endeavor of the firm.
43	Taylor, J., and Vithayathil, J. 2018. "Who Delivers the Bigger Bang for the Buck: CMO or CIO?," *The Journal of Strategic Information Systems* (27:3), pp. 207-220.	CIOs; TMT	Position	Technology leader represented in top management team; marketing leader represented in top management team	Total firm output 1 year lag; total firm output 2 year lag; total firm output 3 year lag; total firm output 4 year lag; total firm output 5 year lag	Null	• Data from Compustat and Execucomp databases(January 1992 to January 2017). • A powerful technology leader can do better in the prediction of a sustained increase ver a one-year, five-year, and seven-year period in firm output than a powerful marketing leader.
44	Gonzalez, P. A., Ashworth, L., and McKeen, J. 2019. "The CIO Stereotype: Content, Bias, and Impact," *The Journal of Strategic Information Systems* (28:1), pp. 83-99.	CIO	Position	CIO characteristics	Student qualification; student punitive action; senior manager qualification; senior manager punitive action	Stereotype theory	• Study1: Trait rating study; study2&3: experiment. • CIOs have more in common with IT personnel than leaders. The CIO stereotype causes the biasing perceptions of CIOs' suitability to occupy strategic roles (i.e., a "glass ceiling" effect) and the extent to which they are blamed for negative outcomes (i.e., a "glass cliff" effect).

Appendix C

Summary of Extant IS Research that Related to IT Orientation

No.	Source	Category	Focus	IV (Mediator/Moderator)	Theory	Relevant Key Findings
1	Banker, R., Hu, N., and Pavlou, P. 2006. "IT Orientation, CIO Reporting Structure, and Firm Performance: To Whom Should the CIO Report, University of California, Riverside–Singapore Management University	Top management	Examines the CIO reporting structure and the alignment with the firms' strategy.	CIO reporting structure	General theoretical underpinning	• IT orientation is framed as a general term that describes the firm's main purpose in using IT. • Firms have two critical types of IT orientation: strategic vs. operational. • In this paper, the firm's IT orientation (automate or Information) is a dummy variable that captures the strategic role of IT in an industry (Information Week's industry designations).
2	Sobol, M. G., and Klein, G. 2009. "Relation of CIO Background, IT Infrastructure, and Economic Performance," *Information and Management* (46:5), pp. 271-278.	CIO; top management	Position	CIO characteristics (age, gender, education, technical position); IT orientation (utilitarian, strategic); IT infrastructure (technical services, application services, management services, hardware, data services)	Upper echelons theory	• The IT orientation of the firm was considered as either strategic, encouraging innovative approaches of IT to support long-term goals or as a utilitarian, emphasizing rigid support of operations. • The IT orientation was determined using a scale developed by Weill and Broadbent (1998).
3	Marino, A. J. 2009. "The Effectiveness of Chief Information Officers in Companies of Central Florida as Perceived by Top-Management-Team Members: An Exploratory Study," Doctoral Dissertation, Capella University.	Top management	Position	Business knowledge; Leadership ability; communication effectiveness; effectiveness of the reporting level	The information-technology function; the top management team	• Data from three research interviews (a) fully structured, (b) semistructured, and (c) unstructured. • Although CIOs do not establish the overall strategies of their organizations, related literature encourages these professionals to develop a general business orientation to supplement their innate IT orientation.
4	Kowshik, Raghu V. 2010. "A Structural Equation Model of the Factors Associated with Influence and Power of IT Departments and Their Relationship to Firm's IT Orientation and Business Performance," Doctoral Dissertation, Walden University.	Business performance	IS planning	Accountability; innovativeness; customer connectedness; partnering with other departments, IT department's influence and power; IT orientation	Resource dependence theory; strategic contingency theory of intraorganizational power; information as a source of power	• IT orientation is defined as a firm-level capability as a business culture that (a) practices the pervasive use of information communication technologies to maintain high levels of business performance, and (b) provides strong top management support to use information technologies. • IT orientation has three characteristics: (1) IT orientation is the pervasive use of information technologies for communication; (2) constant enthusiasm to exploit every opportunity to convert the manual transaction into electronic transactions; (3) eExtensive use of Internet technology.
5	Masli, A., Richardson, V. J., Sanchez, J. M., and Smith, R. E. 2011. "The Business Value of IT: A Synthesis and Framework of Archival Research," *Journal of Information Systems* (25:2), pp. 81-116.	Business performance	IT value	IT investment; IT capability; IT orientation; business leadership IT complementary; alignment of IT with business strategy	Null	• Review papers. • IT orientation means the role of IT in the business and industry. • IT orientation is the fundamental pillar that supports the IT initiative, IT investment, IT capability, and IT performance in firms.

No.	Source	Category	Focus	IV (Mediator/Moderator)	Theory	Relevant Key Findings
6	Thatcher, J., Dinger, M., and George, J. F. 2012. "Information Technology Worker Recruitment: An Empirical Examination of Entry-Level IT Job Seekers' Labor Market," *Communications of the Association for Information Systems* (31:1).	IT recruitment	Positions	IT orientation; job and firm characteristics; size preferences; self esteem; gender and GPA; job seekers and job search behavior	Signaling theory and image theory	• Survey data from 491 entry-level IT job seekers and 412 firm recruiters. • IT orientation refers to the role of information technology in a firm.
7	Karpovsky, A., and Galliers, R. 2012. "Sources of Power and CIO Influence and Their Impact: An Explorative Survey," in *Proceedings of the 34th International Conference on Information Systems*, Orlando, pp. 165-176.	Top management	IS strategic planning, CIO, power, influence, top management, survey	Bureaucratic; network; critical contingencies; psychological	Null	• Survey data from 73 CIOs. • Cash et al. (1992) suggest a reciprocal relationship with the firm's IT orientation and CIO reporting structure. In a strategically IT-oriented firm, the CIO is often a member of the TMT and involved in the firm's strategic planning. However, in an operational IT orientation firm, the CIO's role is limited to heading the IT function, offering IT support, and managing less risky IT projects. • The IT orientation is categorized as either strategic or operational of how the firm treats the role of IT in their business.
8	Al-Taie, M. Z., Lane, M., and Cater-Steel, A. 2014. "The Relationship Between Organizational Strategic IT Vision and CIO Roles: One Size Does Not Fit All," *Australasian Journal of Information Systems* (18:2).	Top management	Examines the impact of vision on the CIO role and structural power	Strategic IT vision	Contingency theory	• This paper argues that IT orientation is a synonym of IT vision, which refers to the shared, aspired state of the role that IT should play in the firm. • There is a positive relationship between an organization's IT vision and the CIO's structural power.
9	Vij, S., and Farooq, R. 2016. "Moderating Effect of Firm Size on the Relationship Between IT Orientation and Business Performancem" *IUP Journal of Knowledge Management* (14:4).	Business performance	Examines the moderating role of firm size in the relationship between IT orientation and business performance.	IT orientation; firm size	Resource-based view (RBV) of the firm	• The survey questionnaire was administered on 240 managerial level employees (CXO's, top-level, and middle-level managers who were critical decision-makers in the organizations). • Information technology orientation is the firm's capability to manage and use information effectively. • The findings provide support for a significant positive relationship between IT orientation and business performance. • The study also provides empirical evidence that firm size moderates the relationship between IT orientation and business performance.

No.	Source	Category	Focus	IV (Mediator/Moderator)	Theory	Relevant Key Findings
10	Andrade Rojas, M. G., Saldanha, T., Khuntia, J., and Kathuria, A. 2016. "IT Orientation Effects on Obstacles and Facilitators of Innovation: An Emerging Economy Perspective in Mexico," in *Proceedings of the 22nd Americas Conference on Information Systems*, San Diego.	Business performance	Innovation, IT orientation, emerging economy, obstacles	Innovation impact; internal obstacles; economic obstacles; policy obstacles; economic facilitators; policy facilitators; internal IT orientation to innovation; external IT orientation to innovation; firm size; staff dedicated to innovation; creativity federal/state; support for innovation; industry sector	Null	• Survey data from 394 firms in Mexico. • IT orientation to innovation is the orientation of a firm's IT function toward innovation.
11	Diesveld, M. 2018. "IT Orientation: What it Comprises and How it Directly Influences Firm Performance," Master's Thesis, Radbout University.	Business performance	Firm performance	Business intelligence; IT system configuration; IT management; digital marketing; electronic customer relationship management; firm performance; number of employees; year of foundation; respondent qualification; industry; production/services; self-conducting or outsourcing	Strategic orientations; competence-based theories	• Survey data from qualitative research. • IT orientation positively influences firms performance IT orientation comprises six IT capabilities: (1) business intelligence, (2) IT system configuration, (3) IT management, (4) digital marketing and sales, (5) social, mobile platform management, (6) online customer service. • IT orientation is equal to IT capabilities.
12	Raymond, L., Bergeron, F., Croteau, A. M., and Uwizeyemungu, S. 2019. "Determinants and Outcomes of IT Governance in Manufacturing SMEs: A Strategic IT Management Perspective," *International Journal of Accounting Information Systems* (35).	IT governance	IT performance	Environmental uncertainty; strategic IT orientation; ITG capabilities; IT alignment capabilities	Null	• Survey data from 223 manufacturing SME. • IT governance in SMEs has empirically accosted with IT performance by IT alignment. • IT governance is affected by the SMEs' environmental uncertainty and IT strategy.
13	Al-Surmi, A., Cao, G., and Duan, Y. 2020. "The Impact of Aligning Business, IT, and Marketing Strategies on Firm Performance," *Industrial Marketing Management* (84), pp. 39-49.	Business performance	Examines the alignment of firms' business strategic orientation, IT strategic orientation, and strategic marketing orientation	Business strategic orientation; IT strategic orientation; marketing strategic orientation	Contingency theory; configuration theory	• In this paper, the IT strategy and IT orientation are used interchangeably. • IT strategy (orientation) determines how IT will be used to facilitate electronic communication to support business processes and needs.
14	Mohdzain, Z. 2003. "Information Systems Strategic Planning in Multinationals: From the Subsidiaries' Perspective," unpublished Doctoral Dissertation, Cranfield University.	Information systems strategic planning (ISSP)	IT performance	Subsidiaries' business orientation; responsibility for IS planning; focus of IS planning; IS planning approach; perceived success (subsidiary IT, subsidiary business)	Null	• Field study methods with structured interview. • IS/IT orientation means the responsibility of IS/IT and the strategic focus of IS/IT.
15	Diesveld, M. 2019. "The Role of IT in Creating Customer Value," Master's Thesis, Radboud University	Business performance	Market orientation, IT orientation, strategic orientation, customer value, capabilities, dynamic capabilities	Market orientation; IT orientation; firm performance; firm size; firm age; respondent qualification; production/services	Resource-based view and dynamic capabilities theory	• Data from cold acquisition and personal connections. • IT orientation is a subset of the strategic orientation of firms. The other is a market orientation. • IT orientation also refers to IT strategy. • IT orientation is defined as the degree to which firms focus on business intelligence, IT system configuration, IT management, digital marketing, and sales, social and mobile platform management, and online customer service in their business operations.